Allocating Misallocation: Decomposing Measures of Aggregate Allocative Efficiency

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Abstract

We explore sources of measured misallocation using establishment data from U.S. manufacturing industries. We decompose standard revenue productivity dispersion statistics into contributions by dispersion in revenue margins over costs and dispersion in input cost shares across plants. We establish a formal link between these components and measured allocative efficiency. The results indicate the components contribute similarly to apparent rising misallocation in US manufacturing. We use the mapping between distortions that influence these distinct components to explore the relationship between inferred distortions and mechanisms that influence one or both sources of revenue productivity dispersion. Finally, we show rising misallocation in the US manufacturing sector in the last several decades is pervasive, and yet a few industries account for over half of the aggregate decline.

Keyword: Misallocation, Productivity Dispersion, Allocative Efficiency

JEL Classification: D24, E23, L16

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1 Introduction

The distribution of firms in the United States economy exhibited large and increasing dispersion in revenue productivity in recent decades, both in terms of labor and total factor productivity.[1] Under a strong, but standard set of assumptions in the macro literature, such dispersion is often interpreted as an indication of distortions, and thus captures deviations of an economy away from its efficient frontier (Hsieh and Klenow, 2009; Bils, Klenow, and Ruane, 2021). Nonetheless, this interpretation and its welfare implications depend on the underlying mechanisms driving dispersion. This paper sheds light on the potential for several mechanisms to explain observed productivity dispersion and measured misallocation patterns. We begin by establishing new facts that connect variation in revenue productivity and misallocation with two key concepts: revenue margins over total cost and input cost share dispersion. Since at least Hsieh and Klenow (2009), it has been recognized that distortions could impact either the scale of a business operation or its factor mix or both. We propose decompositions that separately identify the contribution of heterogeneity in revenue margins in total costs (often called margins for short) and heterogeneity in input cost share dispersion (often called mix for short) to both revenue productivity dispersion and measured misallocation. A critical feature of our analysis is the introduction of a mapping of specific distortions into the the empirical margin component vs the mix component.

Using our novel empirical decomposition, we present evidence that rising dispersion in revenue productivity dispersion in US manufacturing is accounted for by increases in both margin and mix components. Moreover, we find that the correlation between these components is increasing over time. In a standard model of misallocation, rising dispersion of TFPR translates into declining allocative efficiency. We enhance the standard model of misallocation to develop a mapping between our empirical margin and mix components and inferred distortions. This mapping relies on a normalization that does not impact overall measured allocative efficiency, but permits a decomposition of changes in allocative efficiency into distinct mix and margin components. We compare our results to alternative normalizations considered in Hsieh and Klenow (2009) and Bils, Klenow, and Ruane (2021). Counterfactuals are sensitive to the normalization choice, both conceptually and empirically. Changing the normalization alters the implied covariance structure across distortions, which can yield counterintuitive results depending on the underlying mechanisms. We argue our normalization intuitively maps mechanisms onto appropriate distortions and generally yields sensible empirical results, while alternative normalizations provide a less clear conceptual mapping and can generate misleading counterfactuals.

Our empirical decomposition of revenue productivity along with the related structural decomposition of allocative efficiency permits a number of insights. We show that that the contribution of each of these components depends on their relationship with the fundamental factors generating firm heterogeneity in outcomes – what we refer to as the role of technical efficiency, denoted by TFPQ. We show that measures of allocative efficiency depend critically on measured dispersion in TFPQ and in turn the correlation of TFPQ with each of the terms in our decomposition. Given our decomposition, we can provide bounds on the contribution of distortions that map into the mar-

[1] See Syverson (2011), Foster, Grim, et al. (2016), Decker et al. (2020), Blackwood, Foster, et al. (2021). In the international context, see Gopinath et al. (2017).

gin components vs mix components. We then provide additional insight by mapping distortions to concrete examples of mechanisms that create dispersion in revenue productivity—specifically, markups, heterogeneous production elasticities, and heterogeneous input prices (or markdowns).

Finally, we show that most of the decline in allocative efficiency is due to within-industry changes and the role of industry composition is relatively small. The decline in allocative efficiency from rising margin distortions is due to both rising within industry margin distortions and a shift in activity towards high and rising margin distortion industries. Most of the decrease in allocative efficiency from rising mix distortions is within industry. In addition, the decline is disproportionately affected by the lower tail of the distribution in the sense that three narrowly defined industries explain more than half of it. In contrast, the remaining more than 450 industries experienced moderate changes.

The paper proceeds as follows. In the next section, we provide a brief summary of the data we use in our analysis which covers large representative samples of establishment for the US manufacturing sector from 1972-2019. Following this, in section 3 we present a formal empirical decomposition of TFPR into margin and mix components of revenue productivity and present the key empirical patterns from this decomposition. In section 4, we present the key concepts in our model of misallocation, and then implement our structural decomposition linked to our empirical decomposition. Key results using this structural decomposition are presented in section 5. We build on these results in section 6 considering alternative mechanisms that may be at work. We explore related patterns of responsiveness and additional robustness in section 7. The dynamics in the cross-industry distribution of allocative efficiency are presented in section 8. Concluding remarks are in section 9.

2 Data and Measurement

The empirical analysis is based on data from the Collaborative Microproductivity Project (CMP) microdata, which is created using the Annual Survey of Manufactures and Census of Manufactures (ASM/CM) and the Longitudinal Business Database (LBD). CMP microdata contain establishment-level information on inputs and output between 1972-2019. More details about statistical properties of ASM can be found in Cunningham et al. (2023), here we only note that in Census years we use information about plants that were included in the ASM, and calculate sample-weighted statistics where appropriate. We use CMP microdata to construct measures of revenue and inputs. We use industry-level deflators for all nominal expenditures of revenue and inputs. TFPR is calculated as the ratio of deflated revenues (adjusted for inventories) and a composite input index using factor elasticities from industry level cost shares. We permit the latter to vary by year. Further discussion of factor elasticities measured in this way and alternatives is provided below. Inputs include capital, labor, materials and energy. Capital stocks are calculated combining book values and investment flows using the perpetual inventory method. Similar to Hsieh and Klenow (2009) we use labor costs to measure the labor input divided by an industry deflator. This reflects a crude way of adjusting for labor quality.[2] Materials and energy input

[2] As a robustness check, we also consider the sensitivity of our results to measuring the labor input using total hours.

expenditures are measured as nominal expenditures deflated with industry level deflators.[3]

3 Empirical Decomposition

Since revenue productivity is a key concept in the present context, it is useful to start with its definition. Revenue productivity, or TFPR, is defined as the ratio of deflated revenue (adjusted for inventories) and the Cobb-Douglas input index for plant i in industry s at time t: $\widehat{R}_{ist}/\Pi_j \widehat{X}_{ijst}^{\alpha_{jst}}$, where factor elasticities α_{jst} are allowed to vary over time but are assumed to be the same across establishments within an industry.[4] That is, $\widehat{R}_{ist}=R_{ist}/P_{st}$ and $\widehat{X}_{ijst}=(w_{ijst}X_{ijst})/w_{jst}$, where w_{jst} denotes an input price index, $w_{ijst}X_{ijst}$ denotes plant i's expenditures on input j, and P_{st} denotes the industry-level output deflator. In this notation, the deviation of log-TFPR from the sectoral TFPR $\widetilde{\text{tfpr}}_{ist}\equiv\widehat{\text{tfpr}}_{ist}\text{-}\widehat{\text{tfpr}}_{st}$ can be written as

$$\widetilde{\text{tfpr}}_{ist} = \underbrace{\ln\left(\frac{R_{ist}}{TC_{ist}}\right) - \ln\left(\frac{R_{st}}{TC_{st}}\right)}_{\text{margin}} + \underbrace{\sum_j \alpha_{jst}\left(\ln\left(\frac{w_{jst}X_{jst}}{TC_{st}}\right) - \ln\left(\frac{w_{ijst}X_{ijst}}{TC_{ist}}\right)\right)}_{\text{mix}}. \tag{1}$$

Equation (1) describes productivity-deviations as a function of two terms. The first one, labeled as "margin", captures how far the margin of revenues over total costs is from the industry average. The second term, labeled "mix", captures the average deviation of input-shares from their respective industry averages. It follows from equation (1) that the variance of $\widetilde{\text{tfpr}}_{ist}$ is given by

$$\sigma^2_{\text{tfpr}} = \sigma^2_{\text{margin}} + \sigma^2_{\text{mix}} + 2\sigma_{\text{mix/margin}}. \tag{2}$$

The first term on the right hand side of equation (2), which we label as "margin"-dispersion, captures variation in the margin of revenue over costs, and is a frequently used measure in the markup literature. The second term captures the cross-establishment variation in cost-shares, which we show is closely linked to variation in the optimal input-mix and refer to it as "mix"-dispersion. The last term denotes the covariance between the two. Figure 1 shows how the standard deviations evolved over time. Consistent with earlier studies, σ_{tfpr} is rising (see Foster, Grim, et al., 2016; De Loecker, Eeckhout, and Unger, 2020; Blackwood, Foster, et al., 2021; Bils, Klenow, and Ruane, 2021; Edmond, Midrigan, and Xu, 2023; Cunningham et al., 2023). Our decomposition shows that a substantial portion of σ_{tfpr} is explained by σ_{margin}. The latter is consistent with evidence of rising margins over total costs used to indirectly measure markups (see, e.g., De Loecker, Eeckhout, and Unger, 2020 and Autor et al., 2020). Interestingly, the increase in σ_{mix} is also non-trivial and the covariance term plays a role, as we will see below.

Figure 2 displays additional facts about these variables. Panel 2a shows the revenue-weighted average of α_{ijst}, which indicate that intermediate inputs (materials) and labor are the most important cost components, and over time the former became more important than the latter. The

[3] Further details on how all relevant variables were calculated can be found in Blackwood, Grim, et al. (2023). CMP microdata files are available to qualified researchers on approved projects in the FSRDC system. We use version 6 of the dataset. We make two adjustments to the CMP sample. First, we only consider plants that have revenue exceed measured total costs. We also trim the data to exclude the top and bottom 1% of the TFPR distribution, as is standard in the literature.

[4] In practice we use cost shares of total costs at the industry level by year for these elasticities.

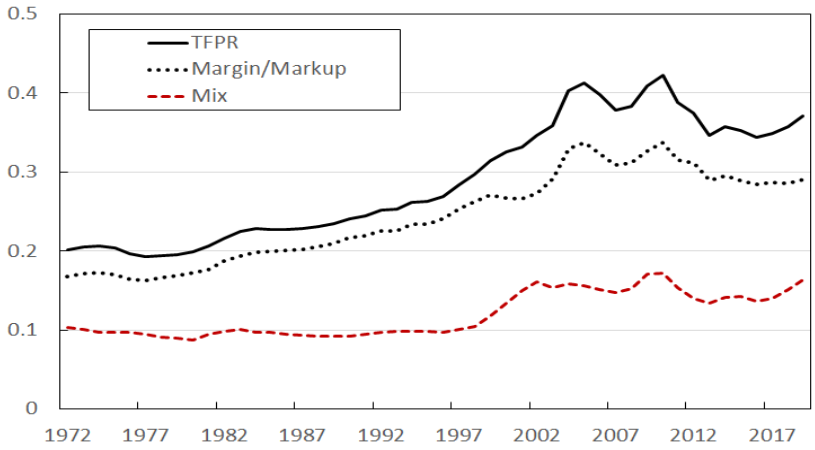


Figure 1: TFPR Dispersion and components

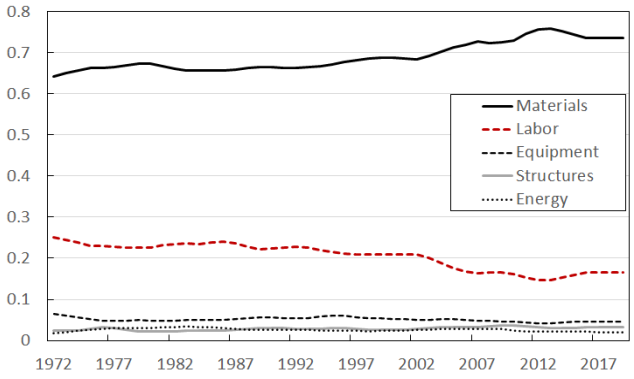
Source: Authors' calculations using CMP data and smoothed with three-year moving average, see Section 2. Each point shows the revenue-weighted standard deviation of $\widetilde{x}_{it}$ where $\widetilde{x}_{it}=x_{ijst}-\overline{x}_{jst}$, and i,j,s,t denote establishments, inputs, industries, and time, respectively.

other shares remained stable. Panel 2b shows the average standard deviation of the components of the mix-term in equation (1). These statistics indicate that dispersion in the shares of intermediate input, labor and energy increased over time, while that of capital did not. Panel 2c shows the margin and mix terms in equation (1) are increasingly correlated. The conclusion from these figures is that both $\sigma^2_{\text{mix,t}}$ and $\sigma_{\text{mix/margin,t}}$ contributed significantly to the rising $\sigma^2_{\text{tfpr,t}}$ in Figure 1. In what follows, we interpret these patterns in the context of a standard model of misallocation, and consider mechanisms that account for them.

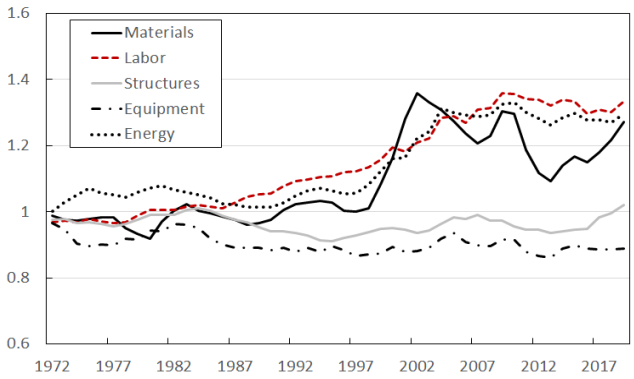
4 A Model of Misallocation

A more formal analysis of allocative efficiency has been developed over the years using a parsimonious modeling framework with simple expressions relating allocative efficiency to plant-level efficiency and distortions. Restuccia and Rogerson (2008) introduced the framework that considers heterogeneous producers with Cobb-Douglas technology, decreasing returns, spot markets for inputs, and idiosyncratic productivities, and focused on how idiosyncratic distortions lead to misallocation and lower aggregate productivity relative to a frictionless case. Consequently, the interpretation is how close (as a percentage) the distribution of inputs is to the efficient allocation. Hsieh and Klenow (2009) extended the model with downward-sloping demand and a constant-returns-to-scale (CRS) production technology and multiple distortion types. Using microdata from the U.S., India, and China they quantify the relative allocative efficiency of these countries. Bils, Klenow, and Ruane (2021) document a decline in allocative efficiency and investigate the role of measurement error using a modified framework with gross output production functions. We follow this methodology to display the decline in U.S. allocative efficiency, shown in Figure 3.[5]

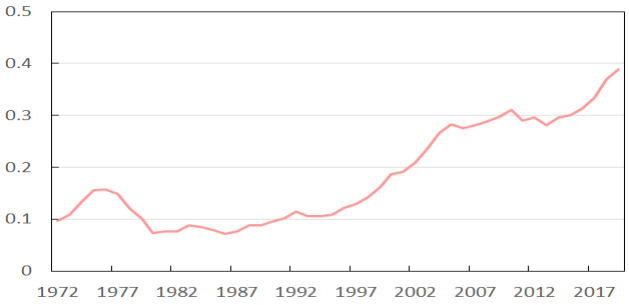
[5]However, as in Blackwood, Foster, et al. (2021), we abstract from the roundabout production model in Bils, Klenow, and Ruane (2021) and treat all inputs as if they are raw materials.



(a) Cost shares



(b) Standard deviation of cost shares, 1972=1



(c) Correlation between mix/margin components

Figure 2: Descriptive statistics of mix/margin components

Source: Authors' calculations using CMP data and smoothed with a three-year moving average, see Section 2. Each point in panel 2a shows the revenue-weighted average of α_{ijst}. Each point in panel 2b shows the revenue weighted-standard deviation of α_{ijst}-α_{jst} normalized to 1972=1. Each point in panel 2c shows the Pearson correlation coefficient between margin and mix components, defined as $\ln\left(\frac{R_{ist}/TC_{ist}}{R_{st}/TC_{st}}\right)$ and $\sum_j \alpha_{jst} \ln\left(\alpha_{jst}/\alpha_{ijst}\right)$ in equation (1), where α_{jst} are estimated as the appropriately weighted average of establishment-level cost shares within each 6-digit NAICS (4-digit SIC).

This figure is broadly consistent with previous findings on substantial declining allocative efficiency in the US manufacturing sector. Comparing Figures 1-3 suggests there is relationship between the variance components and measured allocative efficiency, a relationship that has been established by the above cited papers. We contribute to the literature on allocative efficiency by showing that under a natural normalization σ^2_{margin} is equivalent to variation in an average marginal revenue product term, and σ^2_{mix} is equivalent to variation across inputs in marginal revenue products. We detail this further in the next section.

4.1 Model Setup

The basis of figure 3 and the framework of our analysis is a modified version of the model described in Hsieh and Klenow (2009) and Bils, Klenow, and Ruane (2021), HK-BKR hereafter. All

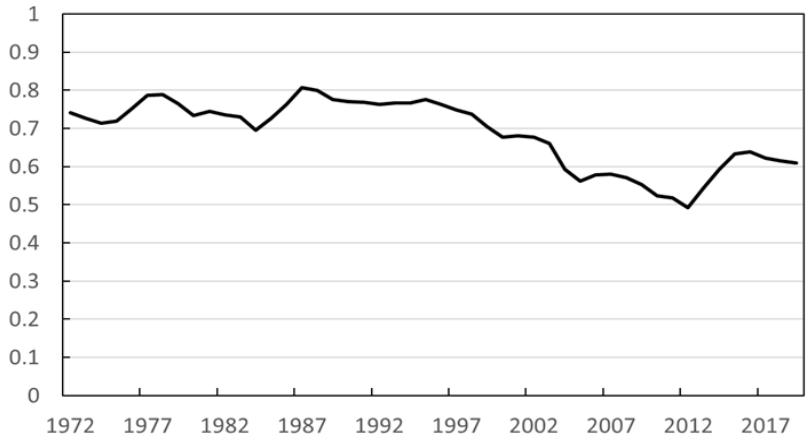


Figure 3: Allocative Efficiency defined as in Hsieh and Klenow (2009)

Source: Authors' calculations using CMP data and a three-year moving average, see Section 2 and Appendix A.2.

production inputs are supplied inelastically.[6] There are no intertemporal decisions in this model, so the path of capital and other inputs is deterministic, and the representative consumer is essentially a passive entity. We suppress time subscripts for simplicity but note that all variables and parameters are time-varying. Total output of the final good is given by the following aggregate of sectoral outputs:

$$Q = \Pi_s Q_s^{\theta_s}.$$

The final firm takes prices as given, which yields the following expression for sectoral prices: $P_s{=}\theta_s(Q/Q_s)$. Normalizing final goods price 1 implies θ_s is defined as the revenue share of sector s. Q_s is specified as a CES-aggregate of varieties i within the sector:

$$Q_s = (\Sigma_i^{N_s} Q_{is}^{\rho_s})^{1/\rho_s}.$$

The sectoral-level producer also takes prices as given implying the following expression for the price of good i: $P_{is}{=}P_s(Q_s/Q_{is})^{1-\rho_s}$. Establishment i's production is given by the following CRS Cobb-Douglas production function across j inputs which are bought/rented on spot markets:

$$Q_{is} = A_{is}\Pi_j X_{ijs}^{\alpha_{js}},$$

where $\Sigma_j \alpha_{js}{=}1$.[7] We use a five-input specification in our empirical implementation: capital equipment, capital structures, labor , energy, and materials. Plants internalize the impact of their production choices on prices, but face distortions both in terms of their inputs and revenues. The production and pricing equations imply the following profit function:

$$\pi_{is} = \max_{X_{ijs}} (1-\tau_{is}^R) P_s Q_s^{1-\rho_s} A_{is}^{\rho_s} \Pi_j X_{ijs}^{\rho_s \alpha_{js}} - \Sigma_j (1+\tau_{ijs}) w_{js} X_{ijs},$$

[6] Alternatively, intermediate goods could be produced in a "roundabout" production network, as in Bils, Klenow, and Ruane (2021). We leave this extension for future research.

[7] BKR postulate roundabout production where output is used as an intermediate good, and Blackwood, Foster, et al. (2021) show this can be extended to the non-CRS case. We maintain the CRS assumption in the empirical analysis but Appendices A.2-A.4 and A.6 are based on a general returns-to-scale parameter denoted by γ_s.

where τ_{is}^R denotes the margin distortion, τ_{ijs} denotes the distortion in input j, and w_{js} denotes the jth input price. The jth first order condition can be written as

$$\rho_s \alpha_{js}(1-\tau_{is}^R)R_{is}/X_{ijs} = w_{js}(1+\tau_{ijs}). \tag{3}$$

The weighted geometric average of the J conditions implies revenue productivity can be written as

$$\frac{R_{is}}{\Pi_j X_{ijs}^{\alpha_{js}}} = \frac{\Pi_j w_{js}^{\alpha_{js}}}{\rho_s \Pi_j \alpha_{js}^{\alpha_{js}}} \frac{\Pi_j (1+\tau_{ijs})^{\alpha_{js}}}{1-\tau_{is}^R}. \tag{4}$$

4.2 Identification and Estimation

In HK-BKR and Blackwood, Foster, et al. (2021), the plant-level composite distortion is given by $\tau_{is} = \Pi_j(1+\tau_{ijs})^{\alpha_{js}}/(1\text{-}\tau_{is}^R)$. Importantly, TFPR_{is} is proportional to τ_{is}, see equation (4). This relationship implies the components on the right hand side of equation (2) are key indicators for allocative efficiency. The structure and solution of these models are well known, and it is generally recognized that while τ_{is} is identifiable from TFPR_{is}, an additional assumption is required to identify revenue and input distortions separately.[8] Thus, any *inferred* revenue or input distortion is dependent upon the normalization, while the total distortion τ_{is} is not.

Our contribution in this context is to show that, under a specific normalization, the mix and margin components in equation (1) are equivalent to the numerator and denominator of τ_{is}, which means that mix and margin distortions are identified by the appropriate cost shares and the revenue-over-total-cost ratio. Appendix A.3 explains how the normalization simplifies first order conditions and establishes the main results in this section. Here we begin with a re-arranged combined first order condition derived therein which expresses revenue over total costs (margin) as a function marginal revenue products $MRPX_{ijs} \equiv \rho_s \alpha_{js} \frac{R_{is}}{w_{js} X_{ijs}}$ and distortions:

$$\frac{R_{is}}{TC_{is}} = \frac{\left(\sum_j \frac{\alpha_{js}}{1+\tau_{ijs}}\right)^{-1}}{\rho_s(1-\tau_{is}^R)} = \frac{1}{\rho_s}\left(\sum_j \frac{\alpha_{js}}{MRPX_{ijs}}\right)^{-1}. \tag{5}$$

Let the following relationship define a normalization of establishment-level inferred input distortions τ_{ijs}:

$$\left(\sum_j \frac{\alpha_{js}}{1+\tau_{ijs}}\right)^{-1} = 1. \tag{6}$$

Equation (6) normalizes input distortions using the harmonic mean. One implication of this normalization is that distortions on revenue are a direct function of profit margins over total cost, or, equivalently, of the (harmonic) average of marginal revenue products:

$$\left(1-\tau_{is}^R\right) = \frac{1}{\rho_s}\frac{1}{\frac{R_{is}}{TC_{is}}} = \frac{1}{\left(\sum_j \frac{\alpha_{js}}{MRPX_{ijs}}\right)^{-1}}. \tag{7}$$

[8] Intuitively, the first order conditions can be used to identify distortions, but there are N_j first order conditions and N_j+1 distortions (N_j input/mix distortions plus one revenue/margin distortion. Hsieh and Klenow (2009) normalize the labor distortion to one, while Bils, Klenow, and Ruane (2021) and Blackwood, Foster, et al. (2021) normalize the margin distortion to one.

Second, inferred input distortions are equivalent to and identified by appropriate cost shares:

$$1 + \tau_{ijs} = \frac{\alpha_{js}}{\alpha_{ijs}}, \tag{8}$$

and further that the inferred input-j distortion is the marginal revenue product of the input normalized by the harmonic mean of marginal revenue products across all inputs:[9]

$$1 + \tau_{ijs} = \frac{MRPX_{ijs}}{\left(\sum_k \frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}}. \tag{9}$$

Another implication is that the margin distortion is identified by the revenue-to-total-cost ratio:

$$\frac{R_{is}}{TC_{is}} = \frac{1}{\rho_s(1 - \tau^R_{is})}. \tag{10}$$

Intuitively, this means that any variation in the (harmonic) mean of marginal revenue products in (5) at the plant-level is "loaded onto" the margin distortion. Empirically, the normalization used here provides a mapping between inferred distortions and the margin and mix components of our empirical decomposition. As we will discuss further below, some specific mechanisms map directly into one or the other of these types of inferred distortions (i.e., we show under simple but commonly used assumptions that markups map directly into τ^R_{is}). In contrast, some mechanisms imply *inferred* distortions impacting both the margin and mix components. Regardless of the underlying mechanisms, our normalization robustly maps variation in the ratio of revenue to total costs onto the margin distortion, and relative variation of input cost shares onto the mix distortion.

We note that the normalization we use to implement this mapping is distinct from that used in Hsieh and Klenow (2009) and in Bils, Klenow, and Ruane (2021). In Hsieh and Klenow (2009), the normalization is that one of the factor specific distortions is zero. In Bils, Klenow, and Ruane (2021) the normalization is the margin distortion is zero. We recognize that the choice of our normalization vs. these alternatives has no implication for overall dispersion in TFPR and allocative efficiency – they are all equivalent. Our normalization has the advantage of decomposing allocative efficiency by mapping revenue to total cost dispersion to margin distortions and relative cost share dispersion to the mix distortions.

Input distortions under the Hsieh and Klenow (2009) normalization yield a similar interpretation to our mix distortion–they capture variation in marginal products relative to the marginal product of the normalized input rather than an average marginal product–and thus can be said to reflect input mix. However, the revenue distortion will capture both input mix and margin dispersion. Likewise, distortions under the Bils, Klenow, and Ruane (2021) normalization will reflect variation in plant-level and input-level mechanisms. As a result, implementing counterfactual exercises where either individual input or revenue distortions are "shut down" are conceptually less appealing. We explore these issues in more detail in section 5.3.

[9] The overall "mix" distortion $\prod_j (1 + \tau_{ijs})^{\alpha_{js}} = \frac{\prod_j (MRPX_{ijs})^{\alpha_{js}}}{\left(\sum_j \frac{\alpha_{js}}{MRPX_{ijs}}\right)^{-1}}$ is the ratio of the geometric average of input marginal products to the harmonic average. The harmonic mean is always lower than the geometric mean except when there is no dispersion in $MRPX$, so the total mix distortion is bounded from below by 1.

Before we implement our decomposition of allocative efficiency, we note that the analysis requires an estimate of technical efficiency A_{is} (sometimes referred to as $TFPQ_{is}$ in the literature). From the profit function, we have $R_{is} = P_s Q_s^{1-\rho_s} A_{is}^{\rho_s} \prod_j X_{ijs}^{\rho_s \alpha_{js}}$ which implies:

$$A_{is} \propto \frac{R_{is}^{\frac{1}{\rho_s}}}{\prod_j X_{ijs}^{\alpha_{js}}}.$$

Given estimates of ρ_s and α_{js}, this allows for identification of A_{is} up to an industry level constant.[10]

In our analysis, we estimate the output elasticities (α_{js} as noted using input cost shares of total costs at the industry level allowing these to vary by year. We estimate ρ_s using the approach of De Loecker and Warzynski (2012). Further discussion of these issues is in Appendix A.4.

4.3 Sectoral TFP and Allocative Efficiency

Allocative efficiency, denoted by AE_{st}, is defined as the ratio of realized and frictionless sectoral total factor productivity, denoted by A_s and A_s^*, respectively. AE_{st} depends on establishment-level efficiencies A_{is} and distortions τ_{ijs}, τ_{is}^R. Appendix A.2 describes these concepts in more detail, here we only state AE_s without the time index:

$$AE_s = \left(\sum_i \left(\frac{A_{is}}{A_s^*} \right)^{\frac{\rho_s}{1-\rho_s}} \left(\frac{\tau_{is}}{\tau_s} \right)^{\frac{-\rho_s}{1-\rho_s}} \right)^{\frac{1-\rho_s}{\rho_s}}.$$

We now take the above (standard) expression, and consider a decomposition that maps to mix and margin components of TFPR dispersion.[11]

4.3.1 Margin and Mix Counterfactuals

The following identity can be used to isolate the effect of margin-distortions:

$$AE_s = \underbrace{\frac{A_s^{*,\text{margin}}}{A_s^*}}_{AE_s^{\text{margin}}} \times \frac{A_s}{A_s^{*,\text{margin}}}, \tag{11}$$

where AE_s^{margin} measures allocative efficiency in the presence of margin distortions only. To see this, consider the definition of $A_s^{*,\text{margin}}$:

$$A_s^{*,\text{margin}} = \left(\sum_i A_{is}^{\frac{\rho_s}{1-\rho_s}} \left(\frac{1-\tau_s^R}{1-\tau_{is}^R} \right)^{\frac{-\rho_s}{1-\rho_s}} \right)^{\frac{1-\rho_s}{\rho_s}},$$

where margin distortions are explicit, and the previous expression gives margin-distorted sectoral TFP. Then $A_s^{*,\text{margin}}/A_s^*$ captures how much margin-distorted sectoral TFP decreases relative to

[10] The industry-level constant can be controlled for by using industry-year effects or dividing by industry-level terms that are proportional to the same constant.

[11] In what follows, we dispense with the distinction inferred and unobserved distortions, and treat all distortions as if they are observed. Note, that the values of the resulting decompositions will depend on the normalization.

the distortionless maximum, and we can interpret it as allocative efficiency with variation in the mix-distortion switched off.[12] We label this ratio as the margin-component of allocative efficiency:

$$AE_s^{\text{margin}} = \frac{A_s^{*,\text{margin}}}{A_s^*} = \left(\sum_i \left(\frac{A_{is}}{A_s^*} \right)^{\frac{\rho_s}{1-\rho_s}} \left(\frac{1-\tau_s^R}{1-\tau_{is}^R} \right)^{\frac{-\rho_s}{1-\rho_s}} \right)^{\frac{1-\rho_s}{\rho_s}}. \tag{12}$$

The analogue of equation (11) for mix-distortions is given by

$$AE_s = \underbrace{\frac{A_s^{*,\text{mix}}}{A_s^*}}_{AE_s^{\text{mix}}} \times \frac{A_s}{A_s^{*,\text{mix}}},$$

where AE_s^{mix} measures allocative efficiency in the presence of the mix-distortion:

$$AE_s^{\text{mix}} = \frac{A_s^{*,\text{mix}}}{A_s^*} = \left(\sum_i \left(\frac{A_{is}}{A_s^*} \right)^{\frac{\rho_s}{1-\rho_s}} \prod_j \left(\frac{1+\tau_{ijs}}{1+\tau_{js}} \right)^{\frac{-\rho_s \alpha_{js}}{1-\rho_s}} \right)^{\frac{1-\rho_s}{\rho_s}}. \tag{13}$$

Note that AE_s^{mix} and $A_s/A_s^{*,\text{margin}}$ have similar interpretation but they are not equivalent. AE_s^{mix} captures how much mix-distorted TFP decreases relative to the distortionless maximum, while $A_s/A_s^{*,\text{margin}}$ measures how much observed TFP –distorted by margin and mix terms– decreases relative to the already scale-distorted TFP. Both ratios capture the effect of mix distortions but $A_s/A_s^{*,\text{margin}}$ confounds the effects of mix- and margin-distortions. Likewise, AE_s^{margin} isolates the effect of margin distortions, while $A_s/A_s^{*,\text{mix}}$ measures the effect of the joint distribution of distortions. We show in Appendix A.5 that these terms are quantitatively similar.

5 Results for Allocative Efficiency

5.1 Counterfactuals

We begin by displaying baseline allocative effciency in the manufacturing sector. The solid line in Figure 4 shows the well-documented decline in allocative efficiency: it was between 70 and 80 percent until the mid-90's, when it began to fall sharply to about 50% in 2011 before rebounding somewhat to over 60% by 2019. Figure 4 also displays the cross-industry averages of the two counterfactuals described in Section 4.3.1, along with an third measure where dispersion in technical efficiency is collapsed. This is relevant because inefficiency increases if either TFPQ dispersion increases or if TFPQ's covariance with distortions changes.

[12] Shutting down the variation in a subset of distortions changes allocations and requires adjusting sectoral means, which capture the extent to which resources would shift across sectors in response to a change in the distortion structure. This adjustment is unnecessary in the standard counterfactual since distortions are collapsed to the industry mean, and so the inter-industry distribution of inputs is preserved. Appendix A.6 describes the adjustment, which we implement in the empirical analysis. It has a negligible effect, so our focus is on the intra-industry implications. More details about the decomposition and alternative interpretations are available in Appendix A.5.

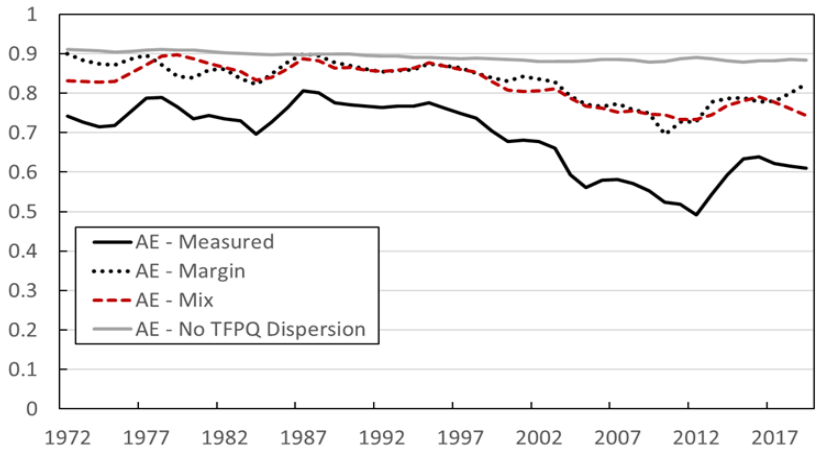


Figure 4: Counterfactural decomposition of allocative efficiency

Source: Authors' calculations using CMP data with a three year-moving average, see Section 2. "AE-Measured" is defined in equation (A6), "AE-Margin" and "AE-Mix" counterfactuals are defined in equations (12) and (13). "AE-No TFPQ Dispersion" is calculated using equation (A6) with the restriction $A_{is}=\widetilde{A}_s$.

The most important patterns from these graphs are as follows. The margin-distortion component of AE_t, denoted by the dotted black line, is about 10 percentage points higher than overall AE prior to 1997. The mix-distortion component, denoted by the red dashed line, follows a similar pattern, which means both distortion types explain roughly the same portion of allocative inefficiency. However, the sum of these contributions does not sum to overall allocative efficiency, although it is generally fairly close. For example, while a three year moving average of overall allocative efficency falls by 23 percentage points from 1997 to 2011, the margin-only counterfactual falls by about 14 percentage points and the mix-only counterfactual falls by roughly 13 percentage points. The third counterfactual, without TFPQ variation (solid gray line), indicates that without TFPQ dispersion allocative efficiency is stable around 90 percent. We can infer from this latter finding that, because shutting down TFPQ variation eliminates the covariance between TFPQ and distortions, the correlation between TFPQ and distortions must play an important role. We explore such corelations in the next section.

5.2 Moments of TFPR and TFPQ

Given the findings of the previous section it is natural to ask whether the observed decline in AE_t is due to increases in distortion dispersion, TFPQ dispersion, or the change in their covariance structure. Despite the fact that the mapping between these descriptive statistics and allocative efficiency is non-trivial,[13] these statistics may be helpful in assessing the relationship between the

[13]Blackwood, Foster, et al. (2021) show what matters for allocative efficiency is the covariance between power functions of TFPQ and distortions. Hsieh and Klenow (2009) note that under the assumption of a jointly log normal distribution of TFPR (with only a revenue distortion) and TFPQ, dispersion in TFPR is sufficient to characterize misallocation without consideration of correlations with TFPQ. We note that the addition of an input distortion in the log normal case means that correlation between distortions influences allocative efficiency as they note in their online appendix. Thus, the rising correlation between mix and margin in Figure 5 has implications for allocative

determinants of allocative efficiency.

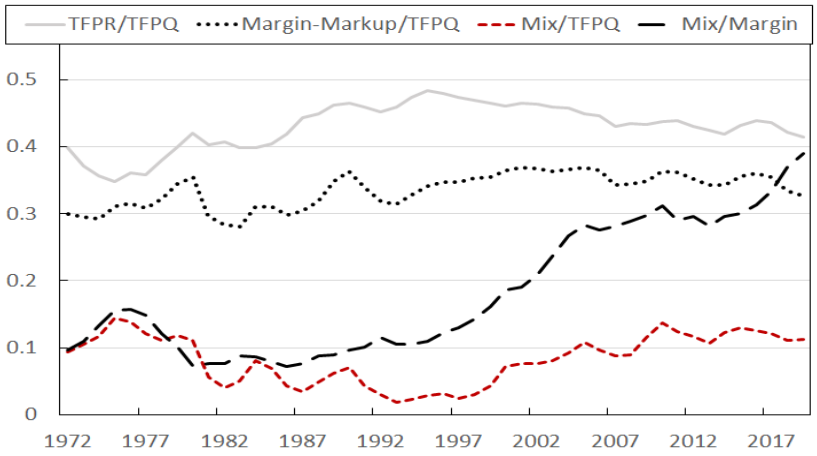


Figure 5: Average correlations between the determinants of allocative efficiency

Source: Authors' calculations using CMP data, see Section 2.

Recall that both TFPR and the margin component showed rising dispersion, the first indicator by roughly 20 log points and margin dispersion by a similar amount, see Figure 1. In addition, dispersion in the mix component remained smaller until the late 1990's, when it began to rise by 5 log points. TFPQ dispersion, not shown here, has also increased substantially since the early 90's. To complete the picture, Figure 5 plots the correlations of these variables. First, TFPR and TFPQ are the most positively correlated (solid gray line) with a value around or above 0.4. That is, more productive plants also tend to be more distorted. TFPQ is positively correlated with the margin component (dotted black line), and the time series pattern mimics that of the composite distortion. The mix component is not as highly correlated with TFPQ (red dashed line). However, the correlation between the mix component and margin component (dashed black line) rose significantly in the 1990's, which roughly coincides with the steepest decline in allocative efficiency. As noted in section 3, the mix and margin components are increasingly correlated, a notable shift underlying the rise in dispersion of TFPR.

5.3 Alternative Normalizations

Our normalization is one of continuum of possible alternatives, but we argue the features of its decompositions recommend it relative to other commonly used normalizations (Hsieh and Klenow, 2009; Bils, Klenow, and Ruane, 2021). To demonstrate, we implement a similar decomposition to our main analysis, but under alternative normalizations used in the literature. Since overall allocative efficiency is the same under these alternatives, we are primarily concerned with counterfactuals where some distortions are shut down. We investigate three alternatives. First, we normalize the materials input, similar to Hsieh and Klenow (2009), but using the largest input share in our gross output production function. Second, we normalize the labor distortion, as in

efficiency, even under log normality. Further, in appendix A.11, we show the tails of the TFPR distribution are sufficiently heavy to suggest log normality does not hold.

Hsieh and Klenow (2009).[14] Under both of these normalizations, we explore counterfactuals where the revenue and input distortions are shut down in turn. Finally, we consider a normalization as in Bils, Klenow, and Ruane (2021). Here, we consider a counterfactual where the materials distortion is eliminated, and one where all other input distortions are eliminated. We abstract from the effect on sectoral distortions (discussed in appendix A.6) to focus on the intra-industry counterfactuals rather than cross-industry distributional effects of shutting down different distortions.

Figure 6 presents the results from these exercises. We find that when a single input distortion is normalized, inference depends heavily on the choice of input. If we normalize the materials input, revenue distortions do most of the work. In fact, removing input distortions actually decreases allocative efficiency when materials are normalized. On the other hand, if we normalize labor, as in Hsieh and Klenow (2009) and panel (c), we see that now removing revenue distortions decreases allocative efficiency. Furthermore, removing either distortion type lowers allocative efficiency in some years. Finally, under the Bils, Klenow, and Ruane (2021) normalization, materials appear to do most of the work, although we note in A.7 this will pick up margin distortions and as well as mix, and in A.8 margin distortions fall disproportionately on the input with the largest cost share (materials).

Before evaluating the merits of alternatives, we first stress the following point: under any normalization, the effect of removing a distortion depends critically on its covariance with other distortions (and TFPQ). Even its sign is dependent on this covariance structure.[15] To see how this is the case for log TFPR dispersion, take the log of the expression in (4) and find the variance of within-industry log TFPR:

$$\sigma^2_{\text{tfpr}} = \sigma^2_{log(\tau_{is})} = \sigma^2_{log\left(1-\tau^R_{is}\right)} + \sigma^2_{log\left(\prod_j (1+\tau_{ijs})^{\alpha_{js}}\right)} - 2Cov\left(log\left(1-\tau^R_{is}\right), log\left(\prod_j (1+\tau_{ijs})^{\alpha_{js}}\right)\right). \tag{14}$$

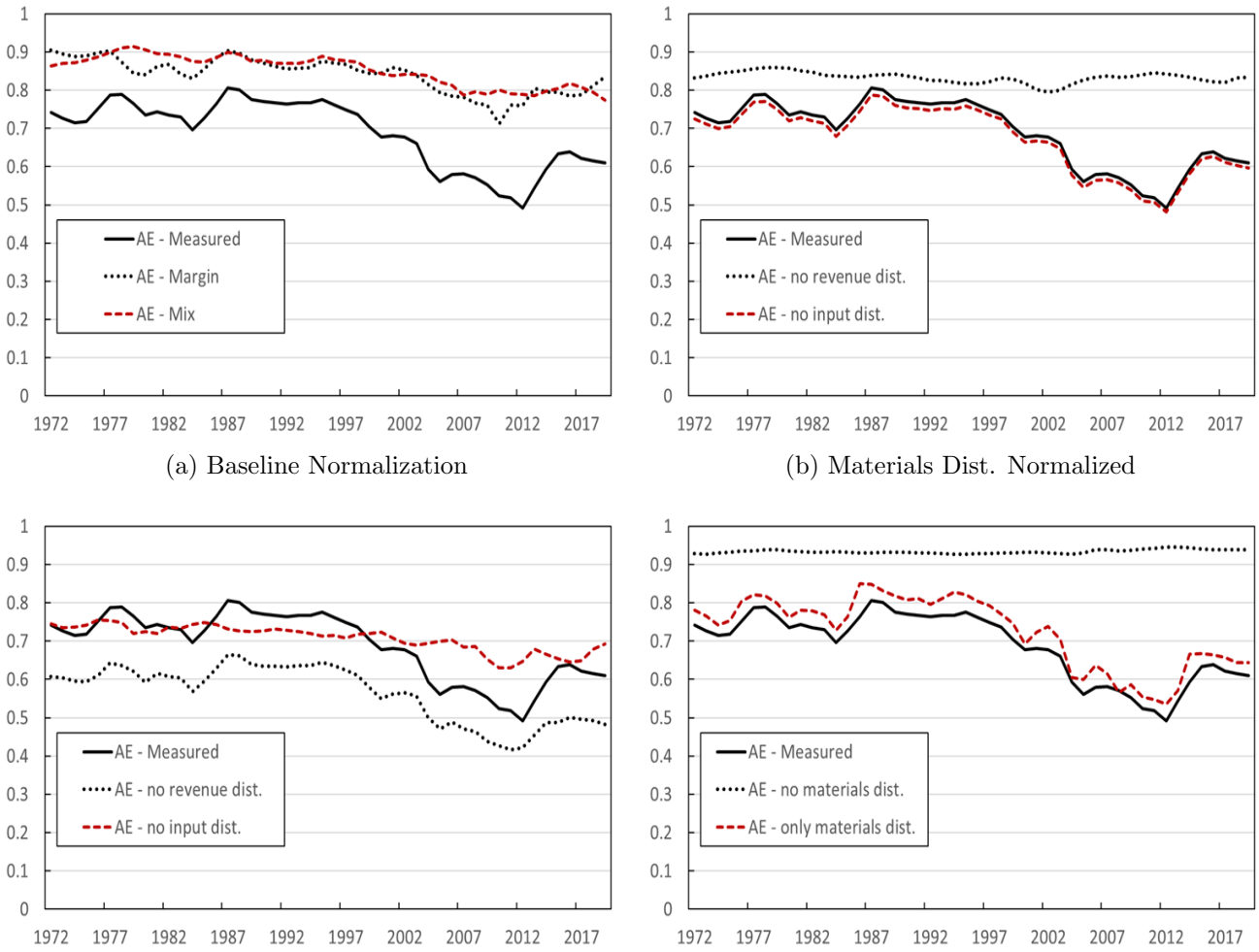
Note that under our normalization, this can be re-expressed as (2) since there is a proportional mapping between input distortions and cost shares in (8) and between revenue distortions and margins in (10). To summarize the contribution of each distortion type, denote $\Delta^{rev}_{\sigma^2_{log(\tau)}}$ as the change in total variance of the plant distortion distribution due to incorporation of revenue distortions, and $\Delta^{mix}_{\sigma^2_{log(\tau)}}$ as the equivalent change due to the incorporation of mix distortions:

$$\Delta^{\text{rev}}_{\sigma^2_{log(\tau)}} = \sigma^2_{log\left(1-\tau^R_{is}\right)} - 2Cov\left(log\left(1-\tau^R_{is}\right), log\left(\prod_j (1+\tau_{ijs})^{\alpha_{js}}\right)\right), \tag{15}$$

$$\Delta^{\text{mix}}_{\sigma^2_{log(\tau)}} = \sigma^2_{log\left(\prod_j (1+\tau_{ijs})^{\alpha_{js}}\right)} - 2Cov\left(log\left(1-\tau^R_{is}\right), log\left(\prod_j (1+\tau_{ijs})^{\alpha_{js}}\right)\right). \tag{16}$$

[14] We note that Hsieh and Klenow (2009) use a value-added production function, so the comparison of their original normalization to our alternatives in a gross output production function framework are not completely analogous. Furthermore, we note that Hsieh and Klenow (2009) do not compute counterfactuals using this normalization, but rather in their appendix consider counterfactuals in a model with only input distortions, as in Bils, Klenow, and Ruane (2021)

[15] These are not entirely new observations. For example, Uras and Wang (2024) show that under a similar normalization to Hsieh and Klenow (2009), removing a subset of distortions can improve allocative efficiency in some industries. Kaymak and Schott (2025) explore how tax heterogeneity can offset other sources of misallocation.



(a) Baseline Normalization

(b) Materials Dist. Normalized

(c) Labor Distortion Normalized

(d) Revenue Distortion Normalized

Figure 6: Counterfactuals under Alternative Normalizations.

Source: Authors' calculations using CMP data, see Section 4.3.1. Panel (a) is our baseline normalization. Panels (b) and (c) are similar to Hsieh and Klenow (2009), with materials and labor normalized, respectively. Panel (d) used the normalization as in Bils, Klenow, and Ruane (2021) where counterfactuals consider removing the materials distortion or all other input distortions. We do not incorporate the effect on sectoral distortions discussed in A.6.

We include the covariance term in both equations since any distortion type influences total dispersion through both its variance and through the covariance term, and likewise, eliminating

any distortion type will also shut down the contribution through covariance.[16] Here it is clear that a positive covariance between the margin term $log(1-\tau_{is}^R)$ and the mix term distortion $log\left(\prod_j (1+\tau_{ijs})^{\alpha_{js}}\right)$ will lower the contribution of both distortion types, and could even flip the sign of the contribution. Note here the negative sign in the first term of the covariance $log\left(1-\tau_{is}^R\right)$, which means an adverse revenue distortion that increases τ_{is}^R lowers $log\left(1-\tau_{is}^R\right)$. So, a positive correlation between $log\left(1-\tau_{is}^R\right)$ and the mix term implies a negative correlation between τ_{is}^R and mix distortions.

Intuitively, to the extent that distortions offset each other, removing a distortion can actually *increase* dispersion and *worsen* allocative efficiency. In appendix A.8, we provide some additional details on how the underlying distribution of marginal revenue products influence these contributions under different normalizations. We show that it is not possible to rule out negative contributions under any normalization, so it is ultimately an empirical question as to whether these contributions are positive or negative for log $TFPR$ variation and measured misallocation. Figures (1) and (4) establish that the contributions work in the expected direction for our normalization and generally produce sensible results, which compare favorably to alternative counterfactuals.

Though we prefer our normalization, which yields intuitive counterfactuals, exploring these alternatives is informative of the underlying structure of the data. The fact that removing some distortions can yield lower allocative efficiency in panels (b) and (c) indicates that revenue distortions and input distortions are correlated under the Hsieh and Klenow (2009) normalization. We explore concrete examples which generate these patterns in the next section. Overall, we show that our normalization generally provides intuitive mappings between mechanisms and distortions, which reinforces our preference for our normalization.

6 Margin vs Mix – Illustrative Examples

The HK-BKR model is heavily used in macroeconomics because of its insights into the connection between micro productivity measures and misallocation and the analytical ease by which productivity differences are mapped into allocative efficiency. The framework points to a set of data moments conveniently computable in typical micro datasets: dispersion statistics of revenue productivity–commonly used in the productivity literature–and the covariance between distortions and TFPQ (see Blackwood, Foster, et al., 2021). As discussed, we contribute by further decomposing TFPR into components of interest–margin and input mix variation.

In this section, we explore illustrative examples of specific mechanisms generating TFPR dis-

[16] Another conceptual variance decomposition which is more appropriate when there are only input distortions, as in Bils, Klenow, and Ruane (2021), is:

$$\sigma_{tfpr}^2 = \sum_j \alpha_{js}^2 \sigma_{log(1+\tau_{ijs})}^2 + 2\sum_j \sum_{k>j} \alpha_{js}\alpha_{ks} Cov\left(log\left(1+\tau_{ijs}\right), log\left(1+\tau_{iks}\right)\right).$$

The appropriate contribution for each input j distortion is then:

$$\Delta^j_{\sigma^2_{log(\tau)}} = \alpha_{js}^2 \sigma_{log(1+\tau_{ijs})}^2 + 2\sum_{j\neq k} \alpha_{js} Cov\left(log\left(1+\tau_{ijs}\right), log\left(1+\tau_{iks}\right)\right).$$

persion, and how they map into the margin vs mix distortions. In some cases, specific mechanisms map cleanly into mix or margin distortions under our normalization. However, in other cases they show up in both. Moreover, we acknowledge that combinations and interactions of mechanisms will further complicate such mappings, and our results from section 5.3 suggest as much. Nevertheless, we emphasize that input cost-share variation resulting from any combination of frictions and distortions impacting input choice will always correspond to our mix distortions, while resulting revenue over total cost dispersion will map onto our margin distortion.

6.1 Markups

A measure of markups used increasingly in the literature emerges from insights from Hall (1988), De Loecker (2011), Antras, Fort, and Tintelnot (2017), De Loecker, Eeckhout, and Unger (2020), and Autor et al. (2020). From the first order condition for variable factors of production, the implied markup is the ratio of the output elasticity to the revenue share of that factor. Under constant returns to scale and treating all factors as variable, the implied markup is the ratio of revenue to total costs. Under these assumptions, the impact of markups is captured completely by the margin distortion τ_{is}^R. Consider the re-arranged the first order condition for the firm with no distortion other than a markup μ_{is}:

$$\rho_s \alpha_{js} \frac{1}{\mu_{is}} R_{is} = w_{js} X_{ijs}.$$

Combining across inputs, we have:

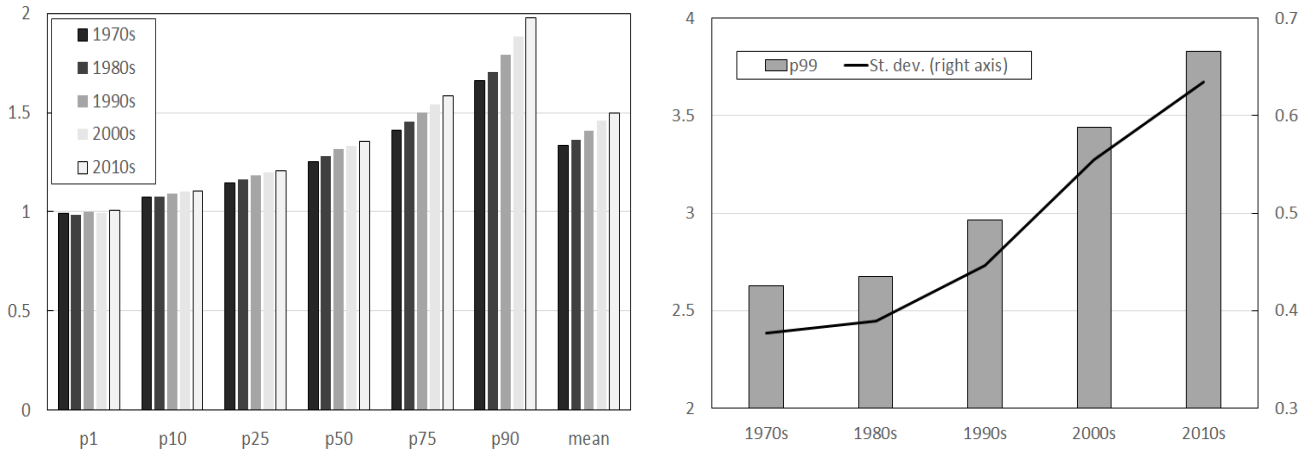
$$\frac{R_{is}}{TC_{is}} = \frac{\mu_{is}}{\rho_s}.$$

This implies:

$$1 - \tau_{is}^R = \frac{1}{\mu_{is}}$$

We can thus interpret the decomposition above as providing a bound on the contribution of markups (so measured) on allocative efficiency. The results above imply that that margin-distortions explain approximately half of the decline. We explore whether the empirical markup distribution shows patterns that can be consistent with these dynamics.

Interpreting the margin distortion as reflecting markups, it is of interest to characterize the distribution of the margin/markup distortion further. Figure 7a depicts the evolution of the markup distribution up to the 90th percentile. These statistics show the smallest changes in the left tail, and somewhat greater changes in the middle of the support of the distribution, with the mean rising from 1.3 to 1.5, a roughly 15 percent increase. In contrast, markups increased significantly in the right tail: 99th percentile rose from about 2.6 to approximately 3.8, which amounts to an almost 50 percent increase, see Figure 7b. Relatedly, the standard deviation (solid line, right axis) increased from below 0.4 to over 0.6, which amounts to a more than 50 percent increase. These findings demonstrate that markup dynamics can be consistent with some of the measured decline in allocative efficiency connected to the rising contribution of the margin component. However, our normalization implies that it cannot be mapped to the mix component. Furthermore, we show in appendix A.9 that labor is becoming less responsive to TFP shocks while



(a) 1st-90th percentiles

(b) 99th percentile and standard deviation

Figure 7: Markup distribution in manufacturing.

Source: Authors' calculations using CMP data, see Section 2.

materials' responsiveness has not declined. While an increase in markups could explain the lower overall responsiveness, it cannot explain the differential trend in responsiveness.

Comparing to alternatives, it is straightforward to show that under the normalization in Hsieh and Klenow (2009) variation in markups across plants will be captured by the revenue distortion. Thus, there is a clean mapping under this alternative normalization as well. Likewise, the Bils, Klenow, and Ruane (2021) normalization provides a mapping between markups and an appropriate average distortion. However, the Bils, Klenow, and Ruane (2021) normalization implies a mapping onto input distortions that is potentially misleading, with the input of the largest share disproportionately affected. In the scenario with only markups, an input with a cost share of 70 percent (like materials) will account for 91 percent of the variation in plant-level distortions. We discuss the relative advantages of our normalization in more detail in A.7 and A.8.

6.2 Heterogeneous Production Elasticities

A potential source of measured mix distortions is that firms may have different production technologies. We show here analytically heterogeneous elasticities in isolation are indeed observationally equivalent to our mix distortions. Furthermore, we show that even when other distortions are present, under our normalization heterogeneity in production elasticities will only be reflected in the mix distortions and not the margin distortion.

The key relationship for identifying distortions is stated in equation (8). For the current analysis, let ε_{ijs} denote the factor by which the elasticity of input j differs from the sectoral cost-share at plant i, i.e. $\varepsilon_{ijs}=\alpha_{js}/\alpha_{ijs}$, and consider a case where this is the only distortion. The modified first order condition is then:

$$\rho_s \alpha_{ijs} \frac{R_{is}}{X_{ijs}} = w_{js}.$$

Re-arranging after multiplying both sides by α_{js}, we have:

$$\underbrace{\rho_s \alpha_{js} \frac{R_{is}}{w_{js} X_{ijs}}}_{MRPX_{ijs}} = \varepsilon_{ijs}.$$

Here, the measured marginal product is equivalent to ε_{ijs}. Our inferred distortion is then

$$1 + \tau_{ijs} = \frac{MRPX_{ijs}}{\left(\sum_k \frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}} = \frac{\varepsilon_{ijs}}{(\sum_k \frac{1}{\varepsilon_{iks}} \alpha_{ks})^{-1}} = \frac{\varepsilon_{ijs}}{(\sum_k \alpha_{iks})^{-1}} = \varepsilon_{ijs}. \tag{17}$$

The last equality in equation (17) makes explicit that input distortions are identical to the respective output elasticity deviations. An important implication is that the mix-distortion discussed earlier is simply a combination of these deviations: $\Pi_j \left(1 + \tau_{ijs}\right)^{\alpha_{js}} = \Pi_j \varepsilon_{ijs}^{\alpha_{js}}$.

The inference about scale distortions is not affected because $(\sum_j \alpha_{js}/(1 + \tau_{ijs}))^{-1} = 1$ implies the relationship between τ_{is}^R and revenues over costs can be written as $(1 - \tau_{is}^R)^{-1} = \rho_s R_{is}/TC_{is}$. This is robust to the inclusion of distortions to output markets like variable markups discussed in 6.1. In this case the re-arranged first order condition is:

$$\underbrace{\rho_s \alpha_{js} \frac{R_{is}}{w_{js} X_{ijs}}}_{MRPX_{ijs}} = \varepsilon_{ijs} \mu_{is}. \tag{18}$$

Then, we have:

$$1 + \tau_{ijs} = \frac{MRPX_{ijs}}{\left(\sum_k \frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}} = \frac{\varepsilon_{ijs}\mu_{is}}{(\sum_k \frac{\alpha_{ks}}{\varepsilon_{iks}\mu_{is}})^{-1}} = \frac{\varepsilon_{ijs}}{\left(\sum_k \frac{\alpha_{ks}}{\varepsilon_{iks}}\right)^{-1}} = \varepsilon_{ijs}. \tag{19}$$

Here, we see that both heterogeneous elasticities are mapped onto mix distortions. Margin distortions are independent of such variation in this case:

$$1 - \tau_{is}^R = \frac{1}{\left(\sum_j \frac{\alpha_{js}}{MRPX_{ijs}}\right)^{-1}} = \frac{1}{\left(\sum_j \frac{\alpha_{js}}{\varepsilon_{ijs}\mu_{is}}\right)^{-1}} = \frac{1}{\mu_{is}\left(\sum_j \alpha_{ijs}\right)^{-1}} = \frac{1}{\mu_{is}}. \tag{20}$$

The conclusion is that under our normalization, variation in elasticities is captured by the mix-distortion, but the measurement of margin distortions is not affected in this case. Comparing to alternatives, we discuss in appendix A.7 the Hsieh and Klenow (2009) normalization will yield a revenue distortion that reflects both the cost-share variation of the normalized input and markups. Likewise, the Bils, Klenow, and Ruane (2021) normalization provides a clean mapping of each of these mechanisms individually, but conflates them when they appear together.

Is this potential source of measured mix distortions a source of allocative efficiency? This is an open question that depends on the source of the dispersion in production technologies. If this dispersion is generated by frictions associated with the adoption of technologies then this is a form of allocative inefficiency. Akcigit and Ates (2021) highlight the role of slow diffusion of adoption of new technologies as a source of misallocation. Alternatively, this may reflect exogenous differences

in production technologies where it is more difficult to make the case that this reflects misallocation. In either case, our decomposition provides an upper bound on this source of measured allocative efficiency as it is captured by the mix component.

An additional complicating issue is the potential impact of heterogeneous production technologies on the measurement of the distribution of TFPQ. Heterogeneous output elasticities likely influence this measurement. In this case, the impact of heterogeneous elasticities on measured misallocation should take that into account.

6.3 Heterogeneous Input Prices

The baseline model assumes that plant's take input prices as given and input prices don't exhibit heterogeneity. However, there may be multiple sources of input price heterogeneity. If such heterogeneity stems from quality differences then the baseline model and measurement approach can be viewed as adjusting for those quality differences. However, such heterogeneity may stem from frictions and distortions in input markets. Geographic market segmentation is likely important in at some input markets (e.g., electricity prices vary across regions given the costs of electricity transmission Davis et al. (2013) and labor markets have a local component). Such market segmentation can also yield market power in input markets. A growing literature (e.g. Card et al., 2018; Lamadon, Mogstad, and Setzler, 2022) highlights the apparent role of markdowns from such market power in labor markets.[17]

In this section we show that a factor specific distortion to prices (such as a markdown) will show up in allocative inefficiency from both margin and mix components. Consider an environment only with a "markdown distortion" on each input j denoted m_{ijs}, the re-arranged plant-level first order condition for labor is given by

$$\underbrace{\rho_s \alpha_{js} \frac{R_{is}}{w_{js} X_{ijs}}}_{MRPX_{ijs}} = m_{ijs}. \tag{21}$$

This implies the following modification of (5):

$$\frac{R_{is}}{TC_{is}} = \frac{1}{\rho_s} \left(\sum_j \frac{\alpha_{js}}{MRPX_{ijs}} \right)^{-1} = \frac{1}{\rho_s} \left(\sum_j \frac{\alpha_{js}}{m_{ijs}} \right)^{-1}.$$

Translating this into the inferred distortions, we have

$$1 - \tau_{is}^R = \frac{1}{\left(\sum_j \frac{\alpha_{js}}{m_{ijs}} \right)^{-1}}. \tag{22}$$

In this equation, input markdowns affect the inferred margin distortion insofar as they affect the average wedge between revenue and costs. In other words, the (harmonic) *average* markdown across inputs acts just like a mark*up* and generates inferred margin distortions.

[17] A related measurement issue is the use of industry-level deflators to infer real quantities. We explore the sensitivity of our results on the labor margin, where we have data on wage bill and hours (and therefore implied price variation), in appendix A.10 and find results are robust.

Likewise, we can show input shares, and thus our inferred input distortion, are affected by the markdown:

$$1+\tau_{ijs} = \frac{m_{ijs}}{\left(\sum_k \frac{\alpha_{ks}}{m_{iks}}\right)^{-1}}, \tag{23}$$

Here, we see that our input distortions capture the relative markdown to the (harmonic) average markdown. In other words, our input distortions reflect variation in markdowns across inputs relative to the mean.[18] We note here (and discuss in more detail in appendix A.7) that variation in markups would not alter this expression, and would solely appear in the expression for the margin distortion.

The total inferred "mix" distortion will be:

$$\prod_j (1+\tau_{ijs})^{\alpha_{js}} = \frac{\prod_j m_{ijs}^{\alpha_{js}}}{\left(\sum_j \frac{\alpha_{js}}{m_{ijs}}\right)^{-1}}. \tag{24}$$

The mix distortion is a ratio of the geometric mean to the harmonic mean in this equation. Since the harmonic mean is bounded from above by the geometric mean, this term is minimized when markdowns are collapsed to 1.

To explore a bit more concretely, take for example the case when a single input k (say, labor) is impacted by markdowns. Our inferred revenue and input distortions are:

$$1-\tau_{is}^R = \frac{1}{\left(1-\alpha_{ks}+\frac{\alpha_{ks}}{m_{iks}}\right)^{-1}},$$

$$1+\tau_{iks} = \frac{m_{iks}}{\left(1-\alpha_{ks}+\frac{\alpha_{ks}}{m_{iks}}\right)^{-1}}, \quad 1+\tau_{ijs} = \frac{1}{\left(1-\alpha_{ks}+\frac{\alpha_{ks}}{m_{iks}}\right)^{-1}} \quad \forall j \neq k$$

Here, we see the markdown contributes to the margin distortion (through the average term) and to the mix distortion. We do not regard this example as implying limitations of our decomposition into margin vs mix distortions. Rather, we think it insightful to recognize that some mechanisms will show up in both components. This is intuitive as higher distortion in some specific factor will induce the firm to be smaller and to substitute away from that factor, and thus impact both margin and mix components of TFPR in our empirical decomposition.

We also note that this same property holds under the normalization of Hsieh and Klenow (2009)–markdowns are reflected in both input and revenue distortions. Furthermore, we show in this case the implied distortions offset each other, and eliminating either distortion *lowers* efficiency. This is consistent with the patterns seen in figure 6 (b) and (c). Under our normalization, we find that up to a first-order, the margin distortion and mix distortions contribute positively to variance when small plants are more distorted. We detail such scenarios in appendix A.7 and A.8, and explore how different mechanisms translate to distortions under different normalizations.

[18] Under the Hsieh and Klenow (2009) normalization, input distortions would capture markdowns relative to the "normalized" input. The Bils, Klenow, and Ruane (2021) normalization appropriately maps the markdown to the input associated with the markdown. We discuss this more in appendix A.7.

7 Industry Analysis

The results so far explore decomposition of aggregate allocative efficiency across input-mix and revenue. However, there may be variation in patterns of allocative efficiency across industries as well. Furthermore, depending on how results are aggregated across industries, some sectors may be contribute disproportionately to overall manufacturing misallocation. We use sectoral decomposition to show that it is both the case that rising misallocation is pervasive, but that a few industries are quantitatively important for aggregate outcomes.

7.1 Industry Decompositions

Log-linearizing AE_t yields $\ln AE_t = \Sigma_s \theta_{st} \ln AE_{st}$, where θ_{st} denotes the revenue share of industry s at time t. Discrete-time changes are given as $\Delta \ln AE_t = \sum_s \theta_{st} \ln AE_{st} - \sum_s \theta_{st-1} \ln AE_{t-1}$. Following Foster, Haltiwanger, and Krizan (2001), we can rewrite $\Delta \ln AE_t$ as

$$\Delta \ln AE_t = \underbrace{\Sigma_s \theta_{st-1} \Delta \ln AE_{st}}_{\text{Within}} + \underbrace{\Sigma_s (\theta_{st} - \theta_{st-1}) \ln AE_{st-1}}_{\text{Between}} + \underbrace{\Sigma_s (\theta_{st} - \theta_{st-1}) \Delta \ln AE_{st}}_{\text{OP Covariance}} \tag{25}$$

Equation (25) parses $\Delta \ln AE_t$ into contributions by within-industry growth, shifts in industry shares, and a term capturing the covariance between growth rates and share changes. It can be used to quantify how much of the overall decline is due to within-industry changes in allocative efficiency and changing industry composition. The components are readily computable for the counterfactuals and individual industries, as well. We use this last property to show in Section 7 that $\Delta \ln AE_t$ is sensitive to atypical industry growth rates.

Figure 8 plots descriptive statistics of the cross-industry distribution of AE_{st}. The 75th percentile varies between 85 and 92 percentage points throughout the entire sample period. There is a modest decline in the median indicating smaller changes in most industries. The 25th percentile decreases substantially after 2000, and AE_t is generally lower than the 1st quartile suggesting disproportionate contributions by low-AE_{st}-industries.

Table 1 summarizes the results of decomposition (25) using data from 1997 and 2011 when AE_t declined the most. The first entries in the first and last columns indicate that approximately 44 of the total decline of 52 log points is due to within-industry changes (85 percent). In contrast, industry-composition explains relatively little of the total decline, although it does appear to contribute indicating some reallocation towards relatively less efficient industries (13 percent, column 2).

The counterfactuals provide guidance on the role of margin distortions vs mix distortions. Only about half of the margin distortion contribution is accounted for by within industry variation. Shifts towards industries with high and rising margin distortions contributes almost as much. For mix distortions, within industry effects dominate.

The rest of the table shows the results for the left-tail of AE_{st}'s growth rate distribution, which is useful given the patterns in Figure 8. We focus on the three most important industry contributors to the overall. In interpreting the decomposition for these industries, the reported statistics reflect the contribution of that industry to the aggregate. The second entry in the last column of Table 1 shows that 19 out of the aggregate 52 log points (approximately 37 percent) is accounted for

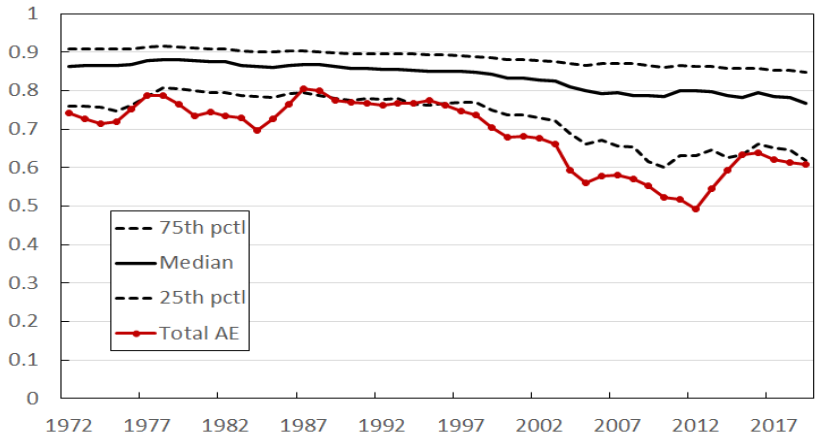


Figure 8: Cross-industry Distribution of AE_{st}

Source: Authors' calculations using CMP data, see Section 2. Note that industry ranks change between time periods and therefore yearly differences in percentiles do not necessary correspond to industry growth rates.

by Petroleum Refineries alone (NAICS 324110). Automobile Manufacturing (NAICS 336111) and Plastics Material and Resin Manufacturing (NAICS 325211) together account for an additional 9.5 log points (next two entries in the last column). These three explain about -29 log points, which amounts to approximately 56 percent of the overall decline. For Petroleum Refineries, only a quarter of the AE_{st}-decline is due to its within contribution to the total (-0.045, column 1), with the rest coming from the between and covariance terms, indicating that this industry is becoming less efficient and more important at the same time. In contrast, the within component is more important in the other two industries: this component contributes more than 100 percent of the overall contribution in Automobiles (-0.122, column 1), and 82 percent in Plastics Manufacturing (-0.018, column 1). Since Automobiles Manufacturing became less important important during this period the negative share- and AE-changes together offset some of the negative effect (column 3). The between and covariance terms in Plastics Manufacturing are negligible.

For the counterfactuals, the margin distortions patterns for the Petroleum Industries exhibit the overall patterns while the mix distortions patterns are small. For the other two industries, only modest contributions arise from the margins distortions alone and mix distortions alone. These findings imply it is the covariance of margin and mix contributions that drive the overall larger contribution for these industries.

Three industries explain 56 percent of the overall decline, while the remaining 44 percent is distributed over more than 450 industries, indicating much milder and fairly broad-based dynamics in the rest of the manufacturing sector. In other words, the dynamics of a single industry may have large effects on AE_t, as illustrated in Figure 9 where we plot AE_t without Petroleum Refineries. Our findings suggest that exploiting the variation in the patterns across industries could shed more light on the sources of rising misallocation in US manufacturing. Furthermore, the sensitivity of aggregate AE_t to outlier sectors, since the aggregate measure is a geometric average across sectors, suggests that aggregation and consideration of sectoral input-output linkages are worth exploring. We leave these matters for future research.

Table 1: $\Delta \ln AE_t$ and its components – Most Important Contributors

	Within Contribution	Between Contribution	OP Covariance Term	Cross-Industry Distortion	Log Change in AE
Total Manufacturing	**−0.444**	**−0.073**	**−0.000**	**0**	**−0.517**
Refineries	−0.045	−0.064	−0.081	–	−0.190
Automobiles	−0.122	0.003	0.046	–	−0.074
Plastic Materials	−0.018	−0.000	−0.004	–	−0.022
Margin Distortions Only					
Total Manufacturing	**−0.150**	**−0.038**	**−0.092**	**−0.041**	**−0.321**
Refineries	−0.053	−0.039	−0.095	–	−0.188
Automobiles	−0.004	0.002	0.000	–	−0.001
Plastic Materials	−0.009	−0.000	−0.002	–	−0.010
Mix Distortions Only					
Total Manufacturing	**−0.115**	**−0.034**	**0.065**	**−0.086**	**−0.170**
Refineries	0.019	−0.033	0.033	–	0.019
Automobiles	0.001	0.001	−0.001	–	0.001
Plastic Materials	−0.005	−0.000	−0.001	–	−0.006

Source: Authors' calculations using equation (25) and CMP data from 1997 and 2011. The column labeled "Cross-Industry Distortion" shows the effect of sectoral distortions, see equation (12) in Blackwood, Foster, et al., 2021.

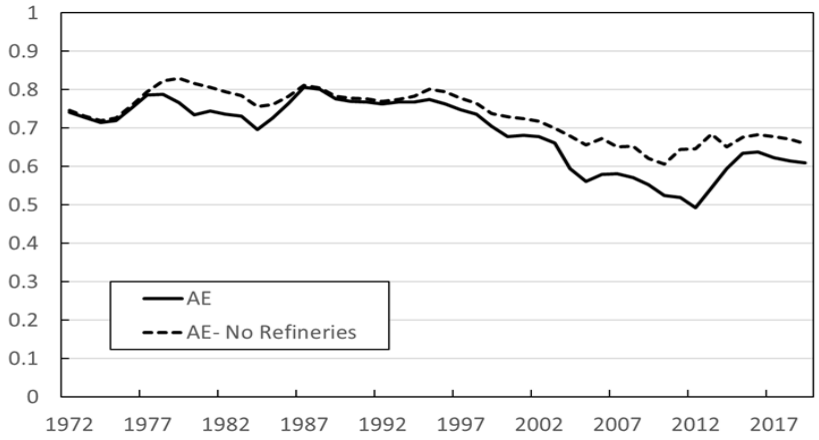


Figure 9: Refineries Contribution to Allocative Efficiency

8 Concluding remarks

We explore the relationship between the productivity distribution and increasing misallocation using establishment-level information from US manufacturing industries. Results indicate that, besides the well-documented rise in dispersion of revenue margins over costs and rising revenue productivity dispersion, the rising dispersion in plant-level input-mix and the correlation between margin- and mix-distortions are also important. These data patterns may be explained by multi-

ple mechanisms. For example, rising margin-dispersion is consistent with market dynamics that lead to consolidation and rising markup differences, while rising mix-dispersion is consistent with increasing differences in the optimal input-mix. The latter can reflect greater heterogeneity in production technologies adopted – which in turn is consistent with the hypothesis of slowing diffusion of new technologies across businesses during this period.

Using the framework described in Hsieh and Klenow (2009) and Bils, Klenow, and Ruane (2021), we show that under a specific normalization the revenue-to-total-cost ratio is equivalent to margin distortions, and that establishment-level cost shares are equivalent to input distortions. We argue that the normalization we use is preferable in that it robustly permits mapping specific distortions readily into either the revenue to total cost dispersion or alternatively input cost share dispersion. For example, measured distortions in the input-mix can be interpreted as indicators of heterogeneous production elasticities, while margin-distortions are more likely to be linked to markups. The normalization allows us to derive counterfactual exercises, which we use to make inference about the importance of the two distortion types. The findings indicate that mix- and margin-distortions are both relevant in explaining the observed decline in allocative efficiency, but neither of them can explain misallocation alone. An interesting part of the overall picture seems to be physical productivity dispersion because without it measured misallocation is very low, which suggests the covariance structure between distortions and efficiency levels is crucial in this context.

Acknowledging that alternative explanations can be mapped into the distortion-types discussed in Hsieh and Klenow (2009) and Bils, Klenow, and Ruane (2021), we review several mechanisms that are consistent with observed data patterns. The discussion focuses on the potentially changing operating environment businesses face. One example is when plants choose different production processes, which could manifest itself as increasing variation in output elasticities. We highlight this point by showing analytically that establishment-level variation in elasticities is isomorphic to input distortions. Alternatively, the dynamics of the estimated markup distribution indicate that factors directly affecting the wedge between revenues and costs, which is equivalent to margin distortions under our proposed normalization, are also relevant. The dispersion in this measure increased significantly, due mostly to dynamics in the right tail. While increased markup variation (consistent with decline in the margin-component of AE_t) could lead to attenuated estimates of responsiveness, the observed changes in the markup distribution do not explain the increase in input mix variation or the differential declines in input responsiveness. In this light, adjustment cost dynamics deserve further exploration as a potential source of variation in optimal input mix and differential responsiveness.

In the last part of the empirical analysis we assess industry dynamics. The data indicate the bulk of the decline in allocative efficiency is due to within-industry changes, and the role of industry composition is less important. Aggregate dynamics are disproportionately affected by the left tail of the distribution in the sense that three narrowly defined industries explain more than half of the decline. The analysis reveals that Cobb-Douglas aggregation yields outsized industry-contributions, shown by the fact that Petroleum Refineries alone account for nearly 40 percent of the allocative efficiency decline, while the rest is distributed over more than 450 industries.

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A Appendix

A.1 Sectoral Distortions: Definition and Alternative Representations

A key component to subsequent analysis will be to adjust, or normalize, individual plant-level distortions based on the appropriate sectoral average. As has been shown in Bils, Klenow, and Ruane (2021), the appropriate mean can be expressed as a revenue-weighted harmonic mean. We show an alternative representation of sectoral distorions in this section. Consider the following equivalent arrangements of the first order condition:[19]

$$\rho_s \alpha_{js} R_{is} \left(1 - \tau_{is}^R\right) = w_{js} X_{ijs} \left(1 + \tau_{ijs}\right)$$
$$\rho_s \alpha_{js} R_{is} (1 + \tau_{ijs})^{-1} = w_{js} X_{ijs} (1 - \tau_{is}^R)^{-1}$$

Multiply appropriately by aggregate variables and then sum across plants:

$$\rho_s \alpha_{js} R_s \left(\sum_i \frac{R_{is}}{R_s} \left(1 - \tau_{is}^R\right) \right) = w_{js} X_{js} \left(\sum_i \frac{X_{ijs}}{X_{js}} \left(1 + \tau_{ijs}\right) \right)$$

$$\rho_s \alpha_{js} R_s \left(\sum_i \frac{R_{is}}{R_s} \frac{1}{1 + \tau_{ijs}} \right) = w_{js} X_{js} \left(\sum_i \frac{X_{ijs}}{X_{js}} \frac{1}{1 - \tau_{is}^R} \right)$$

The expressions in large brackets indicate intuitive definitions for sectoral scale and input distortions. They can be defined as either arithmetic or harmonic means:

$$1 + \tau_{js} = \begin{cases} \sum_i \frac{X_{ijs}}{X_{js}} \left(1 + \tau_{ijs}\right) & \text{Case 1} \\ \left(\sum_i \frac{R_{is}}{R_s} \frac{1}{1 + \tau_{ijs}} \right)^{-1} & \text{Case 2} \end{cases} \tag{A1}$$

$$1 - \tau_s^R = \begin{cases} \sum_i \frac{R_{is}}{R_s} \left(1 - \tau_{is}^R\right) & \text{Case 1} \\ \prod_j \left(\sum_i \frac{X_{ijs}}{X_{js}} \frac{1}{1 - \tau_{is}^R} \right)^{-\alpha_{js}} & \text{Case 2} \end{cases} \tag{A2}$$

Equations (A1)-(A2) show that sectoral input distortions can be written as either input-weighted arithmetic averages or revenue-weighted harmonic averages, and the sectoral scale distortion can be represented a revenue-weighted arithmetic average or as a cost-share-weighed geometric average of input-weighted harmonic means. While Case 1 and Case 2 in equation (A1) can be shown to be equivalent, the cases in (A2) generally are not. Equivalence in (A2) holds under our proposed normalization. Thus, we normalize as in case 1, and summarize the sectoral expression for distortions as follows:

$$\rho_s \alpha_{js} R_s \left(1 - \tau_s^R\right) = w_{js} X_{js} \left(1 + \tau_{js}\right). \tag{A3}$$

We can then show that sectoral average revenue products are proportional to the ratio of these distortions:

$$\rho_s \frac{\alpha_{js}}{w_{js}} \frac{R_s}{X_{js}} = \frac{1 + \tau_{js}}{1 - \tau_s^R}.$$

[19] Edmond, Midrigan, and Xu (2023) make a similar point with regard to the appropriate weighting of markups across plants.

A.2 Sectoral Productivity and Allocative Efficiency

Under the assumptions of the model, aggregate total factor productivity is given by

$$TFP = \frac{Q}{\Pi_j X_j^{\widetilde{\alpha_j}}} = \overline{T} \times \Pi_s A_s^{\theta_s}$$

where X_j is the aggregate amount of input j and $\widetilde{\alpha}_j$ denotes aggregate elasticities. The last expression makes explicit that aggregate TFP can be written as the product of an average distortion $\overline{T}$ and a weighted average of sectoral TFP measures A_s. The latter is defined as:

$$A_s = \frac{Q_s}{\Pi_j X_{js}^{\alpha_{js}}}$$

where $X_{js}{=}\Sigma_i X_{ijs}$ is the sectoral aggregate input j. It can be shown that A_s is a function of establishment-level productivities and distortions:

$$A_s = \left(\sum_i A_{is}^{\frac{\rho_s}{1-\rho_s}} \left(\frac{\tau_{is}}{\tau_s} \right)^{\frac{-\rho_s}{1-\rho_s}} \right)^{\frac{1-\rho_s}{\rho_s}}$$

which is a weighted power sum of plant-level efficiency measures, with power functions of distortions as weights. The exponent in the power sum is determined by the demand-elasticity parameter. Sectoral TFP is maximized if and only if distortions are equalized, i.e. $\tau_{is}{=}\tau_s \ \forall i$, in which case its value is given by

$$A_s^* = \left(\sum_i A_{is}^{\frac{\rho_s}{1-\rho_s}} \right)^{\frac{1-\rho_s}{\rho_s}} \tag{A4}$$

Equation (A4) says the maximized sectoral TFP is a ratio of an unweighted power sum of plants' technical efficiency levels.

Sectoral allocative efficiency is defined as $AE_s{=}A_s/A_s^*$, and $0 \leq AE_s \leq 1$. It represents how much of potential TFP is being realized by a given input allocation. Formally,

$$AE_s = \left(\sum_i \left(\frac{A_{is}}{A_s^*} \right)^{\frac{\rho_s}{1-\rho_s}} \left(\frac{\tau_{is}}{\tau_s} \right)^{\frac{-\rho_s}{1-\rho_s}} \right)^{\frac{1-\rho_s}{\rho_s}}$$

It can be shown that the sectoral allocation of inputs is fixed so long as total supply of the input is in fixed supply, average distortions remain the same, and sectoral wages are equalized (inputs can move freely across sectors), and thus the input index cancels from the previous expression[20], and AE_s can be written as

$$AE_s = \left(\sum_i \left(\frac{A_{is}}{A_s^*} \right)^{\frac{\rho_s}{1-\rho_s}} \left(\frac{\tau_{is}}{\tau_s} \right)^{\frac{-\rho_s}{1-\rho_s}} \right)^{\frac{1-\rho_s}{\rho_s}} \tag{A5}$$

Note that the previous equation holds so long as sectoral distortions are held constant. For the counterfactuals discussed below, sectoral allocations of inputs could change. We account for this effect by calculating the counterfactual input allocation in the NCRS case.

[20] See Blackwood, Foster, et al., 2021 for discussion.

In order to write AE_s as a function of observables, define the sectoral average efficiency as

$$\widetilde{A}_s = \left(\frac{1}{N_s} \sum_i A_{is}^{\frac{\rho_s}{1-\rho_s}} \right)^{\frac{1-\rho_s}{\rho_s}}$$

Multpliying and dividing equation (A5) by N_s, we can rewrite AE_s as a function of $\widetilde{A}_s$:

$$AE_s = \left(\frac{1}{N_s} \sum_i \left(\frac{A_{is}}{\widetilde{A}_s} \right)^{\frac{\rho_s}{1-\rho_s}} \left(\frac{\tau_{is}}{\tau_s} \right)^{\frac{-\rho_s}{1-\rho_s}} \right)^{\frac{1-\rho_s}{\rho_s}} \tag{A6}$$

Equation (A6) can be rearranged to make explicit the role of the covariance between τ_{is} and A_{is}, see equation (12) in Blackwood, Foster, et al. (2021), also discussed discussed in the main text. We use equation (A6) in order to derive the counterfactual AE_s in Appendix A.5.

A.3 Plant-Level normalization

The plant-level first order condition is given by (3). Rearranging and summing over inputs yields

$$\rho_s \left(1 - \tau_{is}^R\right) R_{is} \sum_j \frac{\alpha_{js}}{1 + \tau_{ijs}} = \sum_j w_{js} X_{ijs} = TC_{is}$$

which implies the following expression for the revenue-to-total-cost ratio

$$\frac{R_{is}}{TC_{is}} = \frac{1}{\rho_s} \frac{\left(\sum_j \frac{\alpha_{js}}{1+\tau_{ijs}} \right)^{-1}}{1 - \tau_{is}^R} \tag{A7}$$

Assuming the method described in De Loecker and Warzynski (2012) is used to estimate $\rho_s \gamma_s$, this equation implies deviations from the industry-level markup are due to distortions. However, the system of equations does not identify distortions because the total number of $N_s \times (J+1)$ unknowns (inputs and scale distortions) exceeds the $N_s \times J$ first order conditions. Normalizing $(\Sigma_j \alpha_{js}/(1+\tau_{ijs}))^{-1}=1$ in the previous equation implies

$$\frac{R_{is}}{TC_{is}} = \frac{1}{\rho_s} \frac{1}{1 - \tau_{is}^R} \tag{A8}$$

This means variation in the revenue-to-total-cost identifies the variation in the scale distortion.[21] Combining equations (A7)-(A8) implies

$$1 + \tau_{ijs} = \rho_s (1 - \tau_{is}^R) \alpha_{js} \frac{R_{is}}{w_{js} X_{ijs}} = \frac{TC_{is}}{R_{is}} \alpha_{js} \frac{R_{is}}{w_{js} X_{ijs}} = \frac{\alpha_{js}}{\alpha_{ijs}} \tag{A9}$$

Equation (A9) says that plant-level cost-share variation reflects the variation in the corresponding input distortion. In sum, markup variation identifies the margin distortion, while cost-share variation identifies input distortions because the normalization provides an extra condition for each plant, i.e. the total number of equations equals $N_s \times (J+1)$.

[21] In addition, common movements in input distortions are "loaded onto" the scale distortion.

A.3.1 Effect of normalization on sectoral distortions

Note the following is true based on our assumption:

$$R_{is}\left(1-\tau_{is}^{R}\right)=\frac{TC_{is}}{\rho_s}$$

Summing over plants and dividing both sides by R_s we have:

$$\sum_i \frac{R_{is}}{R_s}\left(1-\tau_{is}^{R}\right)=\frac{TC_s}{\rho_s R_s}$$

Noting that the right hand side is equal to $\left(1-\tau_s^R\right)$ in our first formulation, we have the following after re-arranging:

$$\frac{R_s}{TC_s}=\frac{1}{\rho_s}\frac{1}{1-\tau_s^R}$$

We show in Section A.4 that cost-shares are valid estimates of elasticities if

$$\left(\sum_j \frac{\alpha_{js}}{1+\tau_{js}}\right)^{-1}=1+\tau_{ks}.$$

Furthermore, in order for the methodology in De Loecker and Warzynski (2012) to be valid, $\tau_s^R=\tau_{js}=0$ must hold, as well. The proposed normalization strengthens the condition required by cost-share estimation and DW methodology in the sense that distortions be equalized across inputs and scale distortions, and impose that they be equal to 0 (no sectoral distortions).[22]

A.4 Estimation – Implicit Assumptions

A.4.1 Cost Share Estimation: Equalized Sectoral Factor Distortions

The relevant first order conditions can be summed over plants to obtain the expression for sectoral scale and input distortions in (A3), and then can be rearranged to yield

$$\rho_s\alpha_{js}R_s\frac{1-\tau_s^R}{1+\tau_{js}}=w_{js}X_{js}$$

Summing the previous equation over inputs j we have:

$$\rho_s R_s\left(1-\tau_s^R\right)\sum_j\frac{\alpha_{js}}{1+\tau_{js}}=\sum_j w_{js}X_{js}=TC_s$$

Dividing the former equation by the latter yields:

$$\alpha_{js}\frac{\left(\sum_k\frac{\alpha_{ks}}{1+\tau_{ks}}\right)^{-1}}{1+\tau_{js}}=\frac{w_{js}X_{js}}{TC_s}$$

[22] BKR note that sectoral distortions can be expressed as functions of fundamentals, or distortions and plant-level TFP. We expand on this finding in our paper to properly adjust sectoral distortions when conducting counterfactual exercises.

where the right hand side equals the industry cost-share of input j by definition. The implication is that the condition for cost-shares to be valid estimates of α_{js} is given by

$$\left(\sum_k \frac{\alpha_{ks}}{1+\tau_{ks}}\right)^{-1} = 1+\tau_{js} \quad \forall\, j \tag{A10}$$

For equation (A10) to be true for all inputs, it must be the case that all input distortions are equalized.[23]

A.4.2 DW Methodology: Industry Level Markups to equal to the Weighted Harmonic Mean of Input Distortions

The first order conditions from the previous section can be rearranged as

$$\frac{R_s}{TC_s} = \frac{1}{\rho_s}\frac{\left(\sum_j \frac{\alpha_{js}}{1+\tau_{js}}\right)^{-1}}{1-\tau_s^R}$$

The methodology in De Loecker and Warzynski (2012) estimates markups –technically the markup over variable inputs, but we are assuming all inputs are variable, so the same is true over total costs– by the inverse of ρ_s. The previous equation implies this can only be the case when

$$\left(\sum_j \frac{\alpha_{js}}{1+\tau_{js}}\right)^{-1} = 1-\tau_s^R \tag{A11}$$

Equations (A10)-(A11) imply $1+\tau_{js} = 1-\tau_s^R = \tau_s \;\forall\, j$.

A.5 Counterfactual Decompositions

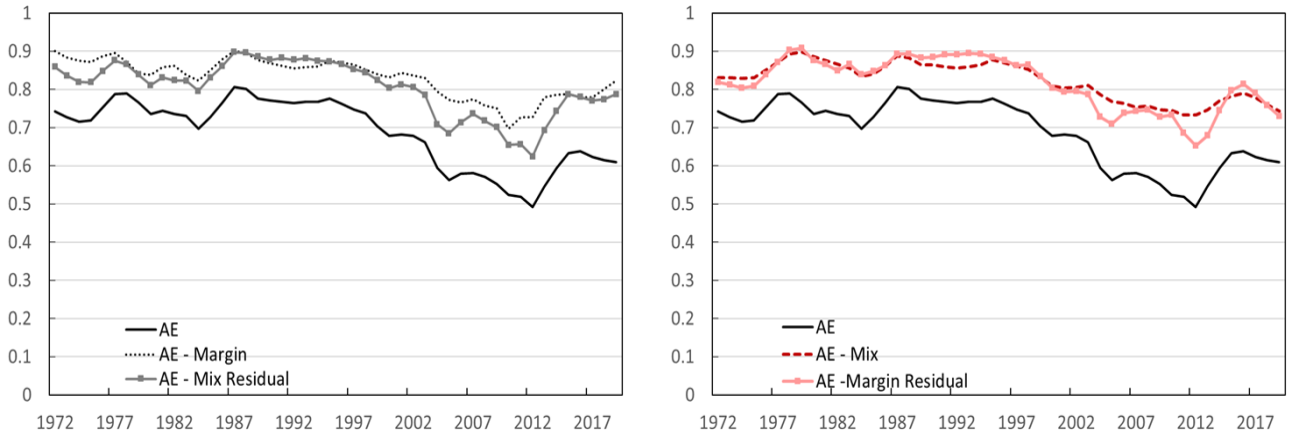
We focus on the first terms of the decompositions in equations (12) and (13) in the main text, which answer the question: "What would allocative efficiency be if there were only margin distortions and mix distortions, respectively?" The first gives us a sense of the first order impact of the scale/margin distortion in isolation, while the second does the same for the input-mix distortions. We thus keep this as our focus as we seek to quantify the direct effects of each distortion type on allocative efficiency.

However, one could also ask "What impact does the additional impact of incorporating mix (or margin) distortions *with margin (mix) distortions already in place*?" In other words, what is the amount of allocative efficiency that margin or mix distortions are *not* explaining in isolation. This is what the second, residual term in equations (12) and (13).

Figure A1 displays the full decompositions of equations (12) and (13), inclusive the additional adjustment to sectoral distortions discussed in A.1. Generally speaking, there appear to be similar contributions of our baseline measure and residual term, which is in line with our conclusion that both distortion types contribute roughly equally to misallocation. However, it does seem that

[23] Suppose not. That is, one distortion is not equal to the others, but is equal to the weighted harmonic average, then it cannot be true that the other distortions are equal to the harmonic average.

residual terms often fall below the main term, especially during the decline of allocative efficiency in the 2000's. Thus, the *combination* of distortion types leads to somewhat larger contributions to allocative efficiency, suggesting the covariance between distortion types is important for explaining changes in allocative efficiency.



(a) AE - Margin and Residual (b) AE - Mix and Residual

Figure A1: Counterfactuals and residual measures: components in equations (11) and (13).

Source: Authors' calculations using CMP data, see Section 4.3.1. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7526913: CBDRB-FY24-CED006-0001, CBDRB-FY24-CED006-0004, CBDRB-FY25-CED006-005, CBDRB-FY25-CED006-015).

A.6 Adjusted Sectoral Distortion Expression

In the counterfactural experiments, when only the variation of certain distortions is collapsed to the mean, other sectoral distortions may be affected. This point can be seen if we combine Case 1 from equation (A1) for τ_{js} (sectoral index is dropped):

$$1+\tau_j = \sum_i \frac{w_j X_{ij}}{w_j X_j}(1+\tau_{ij}) \tag{A12}$$

with the input demand function

$$X_{ij} = \Gamma A_i^{\frac{\rho}{1-\rho}} \frac{\alpha_j}{w_j(1+\tau_{ij})} \prod_k \left(\frac{\alpha_k}{w_k}\right)^{\frac{\rho\alpha_k}{1-\rho}} \left(\frac{1-\tau_i^R}{\Pi_k(1+\tau_{ik})^{\rho\alpha_k}}\right)^{\frac{1}{1-\rho}} \tag{A13}$$

where $\Gamma=(\rho P Q^{1-\rho})^{1/(1-\rho)}$ denotes demand-related sectoral variables. Plugging the expression (A13) into equation (A12), all sectoral multipliers and $(1+\tau_{ij})$ on the right hand side of (A12) cancel and we can write $1+\tau_j$ in terms of exogenous variables and model parameters:

$$\sum_i \frac{\Gamma w_j \frac{\alpha_j}{w_j(1+\tau_{ij})}(1+\tau_{ij}) A_i^{\frac{\rho}{1-\rho}} \prod_k \left(\frac{\alpha_k}{w_k}\right)^{\frac{\rho\alpha_k}{1-\rho}} \left(\frac{1-\tau_i^R}{\Pi_k(1+\tau_{ik})^{\rho\alpha_k}}\right)^{\frac{1}{1-\rho}}}{\sum_i \Gamma w_j \frac{\alpha_j}{w_j(1+\tau_{ij})} A_i^{\frac{\rho}{1-\rho}} \prod_k \left(\frac{\alpha_k}{w_k}\right)^{\frac{\rho\alpha_k}{1-\rho}} \left(\frac{1-\tau_i^R}{\Pi_k(1+\tau_{ik})^{\rho\alpha_k}}\right)^{\frac{1}{1-\rho}}} = \sum_i \frac{\left(\frac{A_i^\rho(1-\tau_i^R)}{\Pi_k(1+\tau_{ik})^{\rho\alpha_k}}\right)^{\frac{1}{1-\rho}}}{\sum_i \frac{1}{1+\tau_{ij}} \left(\frac{A_i^\rho(1-\tau_i^R)}{\Pi_k(1+\tau_{ik})^{\rho\alpha_k}}\right)^{\frac{1}{1-\rho}}}$$

where the right hand side highlights that the jth sectoral distortion depends on the distributions of other distortions. We draw this point out more by rewriting the previous expression using the definition of the plant-level composite distortion $\tau_i = \frac{\Pi_j(1+\tau_{ij})^{\alpha_j}}{1-\tau_i^R}$. Since $\tau_i^\rho = \frac{\Pi_j(1+\tau_{ij})^{\rho\alpha_j}}{(1-\tau_i^R)^\rho} = \frac{\Pi_j(1+\tau_{ij})^{\rho\alpha_j}}{(1-\tau_i^R)(1-\tau_i^R)^{\rho-1}}$, the ratio in the large brackets above can be rewritten as

$$\frac{A_i^\rho(1-\tau_i^R)}{\Pi_k(1+\tau_{ik})^{\rho\alpha_k}} = \frac{A_i^\rho}{\tau_i^\rho}(1-\tau_i^R)^{1-\rho}$$

and the expression for the sectoral distortion is given by

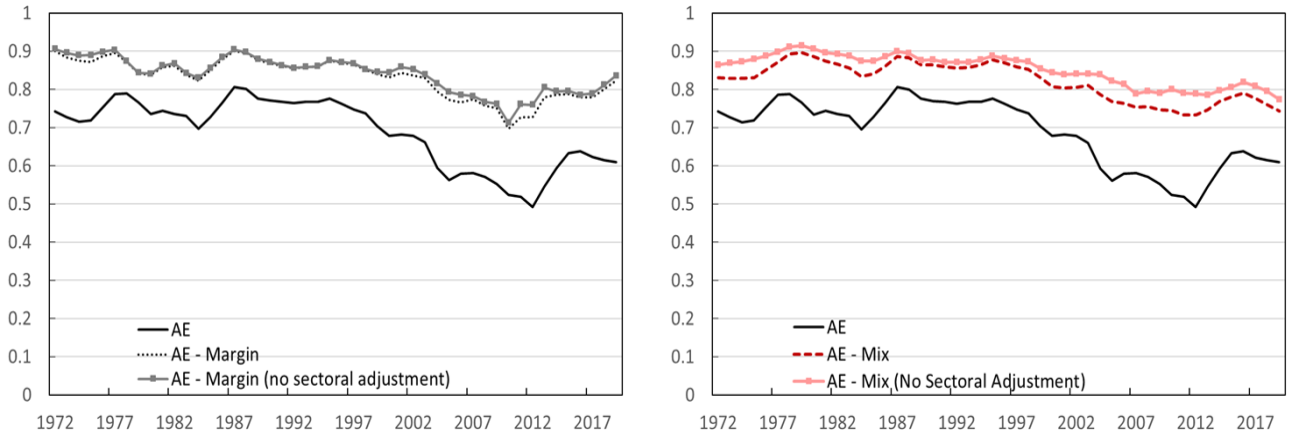
$$1+\tau_j = \frac{\sum_i \left(\frac{A_i}{\tau_i}\right)^{\frac{\rho}{1-\rho}} (1-\tau_i^R)}{\sum_i \left(\frac{A_i}{\tau_i}\right)^{\frac{\rho}{1-\rho}} \frac{1-\tau_i^R}{1+\tau_{ij}}} \tag{A14}$$

Case 1 in equation (A2) and input demand (A13) imply that the sectoral margin distortion is given by

$$1-\tau^R = \frac{\sum_i \left(\frac{A_i}{\tau_i}\right)^{\frac{\rho}{1-\rho}} \left(1-\tau_i^R\right)}{\sum_i \left(\frac{A_i}{\tau_i}\right)^{\frac{\rho}{1-\rho}}} \tag{A15}$$

These expressions highlight that the jth sectoral average not only depends on the jth distribution, but also on the distribution of other input distortions. Changing the distribution of one input changes the sectoral average of other input distortions.

We use equations (A14)-(A15) to adjust sectoral average distortions in counterfactual exercises. To see the effect this term has on our counterfactuals, we display in figure A2 the adjusted and unadjusted measures. Including the sectoral distortions generally adjusts AE downward, but only marginally in the case of the margin counterfactual. The adjustment is more substantial in the case of the mix decomposition, but does not alter our analysis qualitatively.



(a) AE - Margin (b) AE - Mix

Figure A2: Counterfactuals with and without sectoral adjustments.

Source: Authors' calculations using CMP data, see Section 4.3.1. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7526913: CBDRB-FY24-CED006-0001, CBDRB-FY24-CED006-0004, CBDRB-FY25-CED006-005, CBDRB-FY25-CED006-015).

A.7 Inference under Alternative normalizations

In this appendix, we explore how our normalization and the "inferred distortions" from this normalization would capture various patterns in marginal revenue products generated by exemplary mechanisms. In particular we focus on on markups and markdowns that map easily on to distortions. We further explore how our normalization would interpret these differently from other common normalizations in the literature. For simplicity and expositional clarity, we focus on two inputs (labor and capital) as examples.

A.7.1 Markdowns

Suppose the only variation is in MRPL do to a markdown m_{ils} represented in the first order condition for labor:

$$\rho_s \alpha_{ls} R_{is} = m_{ils} w_{ls} L_{is}, \tag{A16}$$

while the capital first order condition is unaffected by the markdown:

$$\rho_s \alpha_{ks} R_{is} = w_{ks} K_{is}.$$

Then, we have:

$$MRPK_{is} = \rho_s \frac{\alpha_{ks}}{w_{ks}} \frac{R_{is}}{K_{is}} = 1 \quad \forall i.$$

$$MRPL_{is} = \rho_s \frac{\alpha_{ls}}{w_{ls}} \frac{R_{is}}{L_{is}} = m_{ils}. \tag{A17}$$

Rearranging and summing over inputs yields

$$\frac{R_{is}}{TC_{is}} = \frac{1}{\rho_s} \left(\frac{\alpha_{ks}}{MRPK_{is}} + \frac{\alpha_{ls}}{MRPL_{is}} \right)^{-1} = \frac{1}{\rho_s} \left(\alpha_{ks} + \frac{\alpha_{ls}}{m_{ils}} \right)^{-1} \tag{A18}$$

Thus the inferred margin distortion

$$\left(1 - \tau_{is}^R\right) = \frac{1}{\left(\alpha_{Ks} + \frac{\alpha_{ls}}{MRPL_{is}}\right)^{-1}} = \frac{1}{\left(\alpha_{Ks} + \frac{\alpha_{ls}}{m_{ils}}\right)^{-1}} \tag{A19}$$

We can see clearly that markdowns will imply "inferred" margin distortions.

The following relationships also hold:

$$\frac{\alpha_{Ls}}{\alpha_{Ks}} \frac{w_{Ks} K_{is}}{w_{Ls} L_{is}} = m_{ils} \tag{A20}$$

$$1 + \tau_{iks} = \frac{\alpha_{Ks}}{\alpha_{iks}} = \frac{MRPK_{is}}{\left(\alpha_{ks} + \frac{\alpha_{ls}}{MRPL_{is}}\right)^{-1}} = \frac{1}{\left(\alpha_{ks} + \frac{\alpha_{ls}}{m_{ils}}\right)^{-1}} \tag{A21}$$

$$1 + \tau_{ils} = \frac{\alpha_{ls}}{\alpha_{ils}} = \frac{MRPL_{is}}{\left(\alpha_{ks} + \frac{\alpha_{ls}}{MRPL_{is}}\right)^{-1}} = \frac{m_{ils}}{\left(\alpha_{ks} + \frac{\alpha_{ls}}{m_{ils}}\right)^{-1}} \tag{A22}$$

Thus, variation in marginal revenue product of labor caused by markdowns alone would translate into both margin variation and mix variation.

How will shutting down these distortions translate to productivity dispersion and allocative efficiency? We show in appendix A.8 the implied approximate contribution, to a first order, for both the contribution of mix and margin distortions can be either positive or negative. However, we have a condition under which the mix distortion contribution to log dispersion is approximately positive: when the average log marginal revenue product is positive.

Note here that the normalization in Hsieh and Klenow (2009) (when the labor distortion is normalized) "loads" any variation in MRPL onto the margin distortion. That is:

$$1 - \tau^{R}_{is,HK} = \frac{w_{ls}}{\rho_s \gamma_s \alpha_{ls}} \frac{L_{is}}{R_{is}} = \frac{1}{MRPL_{is}} = \frac{1}{m_{ils}}. \tag{A23}$$

However, since there is no dispersion in MRPK, this means that $\tau_{ks,HK}$ must cancel out with $\tau^{R}_{is,HK}$, so:

$$1 + \tau_{iks} = \frac{MRPK_{is}}{MRPL_{is}} = \frac{1}{MRPL_{is}} = \frac{1}{m_{ils}} = 1 - \tau^{R}_{is,HK}. \tag{A24}$$

Note here the following possibility. If the above case holds, we can think about counterfactuals as before where we "shut down" each distortion. In this, case, note the total initial distortion is:

$$\tau_{is} = (MRPL_{is})^{\alpha_{ls}} = (MRPL_{is})^{1-\alpha_{ks}} = \frac{(1 + \tau_{iks,HK})^{\alpha_{ks}}}{1 - \tau^{R}_{is,HK}}.$$

The counterfactual that "shuts down" the input distortion yields:

$$\tau^{-K}_{is,HK} = \frac{1}{1 - \tau^{R}_{is,HK}} = MRPL_{is}.$$

Here we see that the removal of the capital distortion actually increases the size of the total wedge, and the resulting log dispersion measure will be amplified. Furthermore, we can see that removing the revenue distortion could also decrease the distortion:

$$\tau^{-REV}_{is,HK} = (1 + \tau_{iks,HK})^{\alpha_{ks}} = (MRPL_{is})^{-\alpha_{ks}},$$

which can have greater log dispersion than $\tau_{is} = (MRPL_{is})^{\alpha_{ls}}$ if $\alpha_{ks} > 0.5$. The upshot here is that removing input distortions potentially leads to *more* counterfactual misallocation than the baseline.[24]

In Bils, Klenow, and Ruane (2021), the margin distortion is normalized to one. In this case, the inferred input distortions are as follows:

$$1 + \tau_{iks,BKR} = 1,$$

$$1 + \tau_{ils,BKR} = MRPL_{is} = m_{ils}.$$

[24]There is a more general tendency for counterfactual allocative efficiency to fall when distortions are correlated. This can be seen, for example, in equation 2 if input mix is negatively correlated with profit margins.

A.7.2 Markups

Consider a case with only a markup distortion μ_{is} which is represented in the FOC as follows:

$$\rho_s \alpha_{js} \frac{1}{\mu_{is}} R_{is} = w_{js} X_{ijs} \tag{A25}$$

We can then derive the following two relationships:

$$MRPX_{ijs} = \mu_{is}$$

$$\frac{R_{is}}{TC_{is}} = \frac{\mu_{is}}{\rho_s}$$

As mentioned in section 6.1, our normalization will appropriately isolate a markup wedge in isolation, since is equal to the inverse distortion. Under the Hsieh and Klenow (2009) normalization, the revenue will also align with the markup wedge:

$$1 - \tau^R_{is,HK} = \frac{1}{MRPL_{is}} = \frac{1}{\mu_{is}}. \tag{A26}$$

Note here that removing the revenue distortion eliminates allocative efficiency. Thus, markups alone cannot be responsible for generating the patterns seen in Figure 6, panels (b) and (c).

Under the Bils, Klenow, and Ruane (2021) assumptions, the geometric average inferred distortions will be isomorphic to the markup, with each input distortion equalized so that $\tau_{ijs,BKR} = \overline{MRPX}$, $\forall j$:

$$\rho_s \frac{R_{is}}{TC_{is}} = \sum_j \left(\frac{\alpha_{js}}{1 + \tau_{ijs,BKR}} \right)^{-1} = \sum_j \left(\frac{\alpha_{js}}{\overline{MRPX}} \right)^{-1} = \overline{MRPX}$$

A.7.3 Markups and Markdowns

However, the above are knife-edge cases. In isolation, either markups and markdowns can be appropriately identified by the HK or BKR normalizations. Yet several issues arise. First, as noted above, counterfactually "shutting down" mix (capital) distortions makes little sense in HK in a world with only labor markdowns, since this necessarily amplifies counterfactual $MRPL$, and therefore $TFPR$, variation. Second, the clean mappings offered by HK and BKR no longer hold in a world where both exist simultaneously. Consider a case with both markup distortions μ_{is} and labor markdown distortions m_{ils} which is represented in the labor first order condition as follows:

$$\rho_s \alpha_{ls} \frac{1}{\mu_{is}} R_{is} = m_{ils} w_{ls} L_{is}, \tag{A27}$$

but not the first order condition for capital, which is only affected by markups:

$$\rho_s \alpha_{ks} \frac{1}{\mu_{is}} R_{is} = w_{ks} K_{is}. \tag{A28}$$

We can then derive the following two relationships:

$$MRPK_{is} = \mu_{is}$$

$$MRPL_{is} = \mu_{is} m_{ils}$$

$$\frac{R_{is}}{TC_{is}} = \frac{1}{\rho_s}\left(\frac{\alpha_{ks}}{MRPK_{is}} + \frac{\alpha_{ls}}{MRPL_{is}}\right)^{-1} = \frac{1}{\rho_s}\left(\frac{\alpha_{ks}}{\mu_{is}} + \frac{\alpha_{ls}}{\mu_{is} m_{ils}}\right)^{-1} = \frac{\mu_{is}}{\rho_s}\left(\alpha_{ks} + \frac{\alpha_{ls}}{m_{ils}}\right)^{-1}$$

Note here that revenue over total costs are no simply a function of the markup, but also the (harmonic) *average* markdown. This makes intuitive sense, in that an average markdown is isomorphic to a markup. Under our normalization, the inferred input distortions will then be:

$$1 + \tau_{iks} = \frac{MRPK_{is}}{\left(\frac{\alpha_{ks}}{MRPK_{is}} + \frac{\alpha_{ls}}{MRPL_{is}}\right)^{-1}} = \frac{\mu_{is}}{\left(\frac{\alpha_{ks}}{\mu_{is}} + \frac{\alpha_{ls}}{\mu_{is} m_{ils}}\right)^{-1}} = \frac{1}{\left(\alpha_{ks} + \frac{\alpha_{ls}}{m_{ils}}\right)^{-1}}, \tag{A29}$$

$$1 + \tau_{ils} = \frac{MRPL_{is}}{\left(\frac{\alpha_{ks}}{MRPK_{is}} + \frac{\alpha_{ls}}{MRPL_{is}}\right)^{-1}} = \frac{\mu_{is} m_{ils}}{\left(\frac{\alpha_{ks}}{\mu_{is}} + \frac{\alpha_{ls}}{\mu_{ils} m_{ils}}\right)^{-1}} = \frac{m_{ils}}{\left(\alpha_{ks} + \frac{\alpha_{ls}}{m_{ils}}\right)^{-1}}, \tag{A30}$$

and our total inferred "mix" distortion will be:

$$(1 + \tau_{ils})^{\alpha_{ls}} (1 + \tau_{iks})^{\alpha_{ks}} = \frac{m_{ils}^{\alpha_{ls}}}{\left(\alpha_{ks} + \frac{\alpha_{ls}}{m_{ils}}\right)^{-1}}. \tag{A31}$$

Thus, input distortions are only a function of the markdown, and capture the relative markdown for an input relative to the (harmonic) average markdown. Furthermore, the overall mix distortion is the ratio of the geometric mean markdown to the harmonic mean markdown, which is bounded from below by 1.

Margin distortions are then given by:

$$1 - \tau_{is}^{R} = \frac{1}{\rho_s \frac{R_{is}}{TC_{is}}} = \frac{1}{\mu_{is}\left(\alpha_{ks} + \frac{\alpha_{ls}}{m_{ils}}\right)^{-1}} \tag{A32}$$

Again, our margin distortion captures both the markup and the (harmonic) average markdown.

In summary, in the presence of both markdowns and markups, our normalization will infer margin distortions as the product of markups and the (harmonic) average markdown. Input distortions will be inferred from input-level markdowns relative to the harmonic mean of markdowns, and will be unaffected by the markup.

Under the Hsieh and Klenow (2009) normalization, distortions are inferred as follows:

$$1 - \tau_{is,HK}^{R} = \frac{1}{MRPL_{is}} = \frac{1}{\mu_{is} m_{ils}} \tag{A33}$$

$$1 + \tau_{iks,HK} = \frac{MRPK_{is}}{MRPL_{is}} = \frac{\mu_{is}}{\mu_{is} m_{ils}} = \frac{1}{m_{ils}} \tag{A34}$$

Here we make two observations. First, the input distortion for capital is still informative about relative markdowns: it is the ratio of the capital markdown (assumed to be one) to the labor markdown. Even if there was a capital markdown, this distortion would still capture the relative value of the markdown compared to labor. Comparing to our normalization, input distortions capture the markdown of an input *relative to the harmonic mean of markdowns.* On the other

hand, the margin distortion no longer maps to revenue over total costs as it does in the markups only case, and is a combination of both markups and labor markdowns.

Under the Bils, Klenow, and Ruane (2021) assumptions, each input distortion would be a combination of markdowns and markups. Assuming the potential for markdowns in both inputs, we have:

$$1 + \tau_{iks,BKR} = MRPK_{is} = \mu_{is} m_{iks}, \tag{A35}$$

$$1 + \tau_{ils,BKR} = MRPL_{is} = \mu_{is} m_{ils}. \tag{A36}$$

This would thus be less informative for distinguishing between the two sources of distortions, although relative markdowns can be obtained by a ratio between the two input distortions. Note, if markdowns in capital markets are nonexistent and capital is otherwise undistorted, as originally assumed, then markups can be backed out from the capital distortion, but this is a knife-edge case.

A.7.4 Heterogeneous production elasticities and Markups

Building off of section 6.2, we note that under heterogenous production elasticities and markups, defining $\varepsilon_{ijs} = \frac{\alpha_{js}}{w_{js} X_{ijs}/TC_{is}}$, $j = k, l$:

$$1 + \tau_{ijs} = \varepsilon_{ijs}, \tag{A37}$$

$$1 - \tau^R_{is} = \frac{1}{\mu_{is}}. \tag{A38}$$

Under the Hsieh and Klenow (2009) normalization, we have:

$$1 + \tau_{iks,HK} = \frac{MRPK_{is}}{MRPL_{is}} = \frac{\varepsilon_{iks}\mu_{is}}{\varepsilon_{ils}\mu_{is}} = \frac{\varepsilon_{iks}}{\varepsilon_{ils}}, \tag{A39}$$

$$1 - \tau^R_{is,HK} = \frac{1}{MRPL_{is}} = \frac{1}{\varepsilon_{ils}\mu_{is}}. \tag{A40}$$

Similar to previous sections, we see that while the recovered input distortions are similar in spirit under both normalizations, the inferred revenue distortion carries a much different interpretation.

Under the Bils, Klenow, and Ruane (2021) normalization, we have:

$$1 + \tau_{iks,BKR} = MRPK_{is} = \mu_{is}\varepsilon_{iks}, \tag{A41}$$

$$1 + \tau_{ils,BKR} = MRPL_{is} = \mu_{is}\varepsilon_{ils}. \tag{A42}$$

Counterfactuals that eliminate these distortions again combine effects of markups and production function heterogeneity.

Finally, we note that while output market distortions like variable markups will be cleanly distinguished from production function heterogeneity in our decomposition, the world may be even more complex. In particular, heterogeneity in production functions might interact with other input distortions. In this case, underlying heterogeneity in the production function may not cleanly map onto mix distortions as before (and likewise, the margin distortion may not be free of the influence of variation in elasticities). However, it remains that the resulting *cost share heterogeneity* from any combination of frictions and distortions impacting input choice will be mapped onto mix distortions.

A.8 Distortion Contributions to TFPR Dispersion

In this appendix, we explore contribution of distortion types under alternative normalizations. In particular, we are interested in the conditions which give rise to negative contributions to log TFPR dispersion. Define $\widetilde{\tau}_{is}^{R} = log(1-\tau_{is}^{R})$ and $\widetilde{\tau}_{is}^{mix} = log\left(\prod_j (1+\tau_{ijs})^{\alpha_{js}}\right)$. Thus, $log(\tau_{is}) = \widetilde{\tau}_{is}^{mix} - \widetilde{\tau}_{is}^{R}$, and we can write from the variance decomposition:

$$\sigma^2_{\text{tfpr}} = \sigma^2_{log(\tau)} = \sigma^2_{\widetilde{\tau}^R} + \sigma^2_{\widetilde{\tau}_{mix}} - 2Cov\left(\widetilde{\tau}_{is}^{R}, \widetilde{\tau}_{is}^{mix}\right).$$

The potential for there to be a net positive contribution (i.e. reduces dispersion) of a distortion type depends on the covariance term. I aim to analyze the potential for covariances to be positive or negative under the HK (2009) assumptions and our assumptions. In these exercises, I will take the values and distribution of the marginal revenue products $MRPX_{ijs}$ as given (they describe the fundamental structure of the data). So, in general I will be looking to reduce relationships between distortions to relationships between marginal revenue products.

First, I note the following define some additional terms and derive an expression for covariance between the two distortion types to be used more generally throughout this analysis. First, define industry average distortions as geometric averages:

$$\tau_s^R = \prod_i \left(1-\tau_{is}^R\right)^{\frac{1}{N}}; \quad \tau_{js} = \prod_i \left(1+\tau_{ijs}\right)^{\frac{1}{N}}; \quad \tau_s^{mix} = \prod_i \left(\prod_j \left(1+\tau_{ijs}\right)^{\alpha_{js}}\right)^{\frac{1}{N}}.$$

Then define

$$\widetilde{\tau}_s^R \equiv log\left(\tau_s^R\right) = \frac{1}{N}\sum_i \widetilde{\tau}_{is}^R; \quad \widetilde{\tau}_s^{mix} \equiv log\left(\tau_s^{mix}\right) = \frac{1}{N}\sum_i \widetilde{\tau}_{is}^{mix}.$$

Now, we can write the covariance between these distortions as follows:

$$Cov\left(\widetilde{\tau}_{is}^{R}, \widetilde{\tau}_{is}^{mix}\right) = \frac{1}{N}\sum_i \left(\widetilde{\tau}_{is}^{mix} - \underbrace{E\left[\widetilde{\tau}_{is}^{mix}\right]}_{\widetilde{\tau}_s^{mix}}\right)\left(\widetilde{\tau}_{is}^{R} - \underbrace{E\left[\widetilde{\tau}_{is}^{R}\right]}_{\frac{1}{\widetilde{\tau}_s^R}}\right) = \frac{1}{N}\sum_i \left(\widetilde{\tau}_{is}^{mix} - \widetilde{\tau}_s^{mix}\right)\left(\widetilde{\tau}_{is}^{R} - \widetilde{\tau}_s^{R}\right).$$

Note the last expression is an average ration of log differences from a mean, so it roughly corresponds to an average ratio of percentage deviations. This can be thought of as a kind of elasticity of one distortion type with respect to the other.

A.8.1 Our Normalization

Recall that under our normalization:

$$1-\tau_{is}^R = \frac{1}{\left(\sum_k \frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}}, \quad 1+\tau_{ijs} = \frac{MRPX_{ijs}}{\left(\sum_k \frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}}.$$

Working along as before, we have the mix distortion

$$\left(1+\tau_{ijs}\right)^{\alpha_{js}} = \frac{\prod_j MRPX_{ijs}^{\alpha_{js}}}{\left(\sum_k \frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}}.$$

Taking logs:

$$\widetilde{\tau}_{is}^{R} = -log\left(\left(\sum_{k}\frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}\right),$$

$$\widetilde{\tau}_{ijs} \equiv log\left(1+\tau_{ijs}\right) = log\left(MRPX_{ijs}\right) - log\left(\left(\sum_{k}\frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}\right)$$

$$\widetilde{\tau}_{is}^{mix} = log\left(\prod_{j} MRPX_{ijs}^{\alpha_{js}}\right) - log\left(\left(\sum_{k}\frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}\right)$$

$$= \sum_{j}\alpha_{js}\left[log\left(MRPX_{ijs}\right) - log\left(\left(\sum_{k}\frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}\right)\right]$$

Applying to the industry average, we have:

$$\widetilde{\tau}_{s}^{R} = -\frac{1}{N}\sum_{i} log\left(\left(\sum_{k}\frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}\right)$$

$$\widetilde{\tau}_{js} \equiv \frac{1}{N}\sum_{i}\widetilde{\tau}_{ijs} = \frac{1}{N}\sum_{i}\left(log\left(MRPX_{ijs}\right) - log\left(\left(\sum_{k}\frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}\right)\right)$$

$$\widetilde{\tau}_{s}^{mix} = \frac{1}{N}\sum_{i}\sum_{j}\alpha_{js}\left[log\left(MRPX_{ijs}\right) - log\left(\left(\sum_{k}\frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}\right)\right] = \frac{1}{N}\sum_{i}\sum_{j}\alpha_{js}\widetilde{\tau}_{ijs}$$

Finally, define the following terms as the log deviations of marginal revenue products from their average:

$$\Delta log\left(\prod_{j} MRPX_{ijs}^{\alpha_{js}}\right) \equiv log\left(\prod_{j} MRPX_{ijs}^{\alpha_{js}}\right) - \frac{1}{N}\sum_{i} log\left(\prod_{j} MRPX_{ijs}^{\alpha_{js}}\right)$$

and

$$\Delta log\left(\overline{MRPX}_{is}\right) \equiv log\left(\left(\sum_{k}\frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}\right) - \frac{1}{N}\sum_{i} log\left(\left(\sum_{k}\frac{\alpha_{ks}}{MRPX_{iks}}\right)^{-1}\right)$$

The covariance term is then:

$$cov\left(\widetilde{\tau}_{is}^{R}, \widetilde{\tau}_{is}^{mix}\right) = -\left(\frac{1}{N}\sum_{i}\Delta log\left(\overline{MRPX}_{is}\right)\Delta log\left(\prod_{j} MRPX_{ijs}^{\alpha_{js}}\right) - \frac{1}{N}\sum_{i}\Delta log\left(\overline{MRPX}_{is}\right)^{2}\right) \tag{A43}$$

The contribution of mix distortions is now:

$$\sigma^{2}_{\widetilde{\tau}_{s}^{mix}} - 2cov(\widetilde{\tau}_{is}^{R}, \widetilde{\tau}_{is}^{mix}) = Var\left(log\left(\prod_{j} MRPX_{ijs}^{\alpha_{js}}\right)\right) - Var\left(log\left(\left(\sum_{j}\frac{\alpha_{js}}{MRPX_{ijs}}\right)^{-1}\right)\right). \tag{A44}$$

The increase in productivity dispersion from adding mix distortions is equivalent to the difference between the variance of the log geometric average of marginal revenue products and the variance of the harmonic average of marginal revenue products. Since this involves the squares of concave functions, it is neither convex nor concave for all positive values, and so cannot be bounded.

For margin distortions, the sign of the contribution depends entirely on the covariance of the harmonic and geometric averages of distortions:

$$\sigma^2_{\widetilde{\tau}_{rev}} - 2cov(\widetilde{\tau}^R_{is}, \widetilde{\tau}^{mix}_{is}) = 2cov\left(log\left(\prod_j MRPX_{ijs}^{\alpha_{js}}\right), log(\overline{MRPX}_{is})\right) - \sigma^2_{log(\overline{MRPX}_s)} \tag{A45}$$

Here, we see the sign of the contribution of the revenue distortion depends on the relative strength of the covariance between geometric means and harmonic means. One way to re-write this is:

$$\sigma^2_{\widetilde{\tau}_{rev}} - 2cov(\widetilde{\tau}^R_{is}\widetilde{\tau}^{mix}_{is})$$

$$= \sigma^2_{\log\left(\overline{MRPX}_s\right)}\left(2\frac{\sigma_{log\left(\prod_j MRPX_{ijs}^{\alpha_{js}}\right)}}{\sigma_{\log\left(\overline{MRPX}_s\right)}} corr\left(log\left(\prod_j MRPX_{ijs}^{\alpha_{js}}\right), log(\overline{MRPX}_{is})\right) - 1\right) \tag{A46}$$

Note that the term $\frac{\sigma_{log\left(\prod_j MRPX_{ijs}^{\alpha_{js}}\right)}}{\sigma_{\log\left(\overline{MRPX}_s\right)}}$ will be greater than 1 if and only if the the mix contribution is positive. So this will tend to be positive if the mix contribution is positive and the correlation between geometric and harmonic means is greater than 0.5.

A.8.2 Hsieh and Klenow (2009) normalization

Without loss of generality, normalize the labor margin.[25] We have the following:

$$1 - \tau^R_{is} = \frac{1}{MRPL_{is}}, \quad 1 + \tau_{ijs} = \frac{MRPX_{ijs}}{MRPL_{is}} \quad \forall j \neq L.$$

This implies we have mix distortion:

$$\prod_j (1 + \tau_{ijs})^{\alpha_{js}} = \frac{\prod_{j \neq L} MRPX_{ijs}^{\alpha_{js}}}{MRPL_{is}^{1-\alpha_{Ls}}}.$$

Taking logs:

$$\widetilde{\tau}^R_{is} = -log\left(MRPL_{is}\right),$$

$$\widetilde{\tau}_{ijs} \equiv log\left(1 + \tau_{ijs}\right) = log\left(MRPX_{ijs}\right) - log\left(MRPL_{is}\right)$$

$$\widetilde{\tau}^{mix}_{is} = log\left(MRPX_{ijs}^{\alpha_{js}}\right) - log\left(\prod_{j \neq L} MRPL_{is}^{\alpha_{js}}\right) = \sum_{j \neq L} \alpha_{js}\left[log\left(MRPX_{ijs}\right) - log\left(MRPL_{is}\right)\right]$$

Applying to the industry average, we have:

$$\widetilde{\tau}^R_s = -\frac{1}{N}\sum_i log\left(MRPL_{is}\right)$$

[25] Note that I use $MRPL_{is} \equiv MRPX_{ijs}$ where $j = L$ is the labor input margin.

$$\widetilde{\tau}_{js} \equiv \frac{1}{N}\sum_i \widetilde{\tau}_{ijs} = \frac{1}{N}\sum_i \left(log\left(MRPX_{ijs}\right) - log\left(MRPL_{is}\right)\right)$$

$$\widetilde{\tau}_s^{mix} = \frac{1}{N}\sum_i \sum_{j\neq L} \alpha_{js}\left[log\left(MRPX_{ijs}\right) - log\left(MRPL_{is}\right)\right] = \frac{1}{N}\sum_i \sum_{j\neq L} \alpha_{js}\widetilde{\tau}_{ijs}$$

Finally, define the following terms as the log deviations of marginal revenue products from their average:

$$\Delta log\left(MRPX_{ijs}\right) \equiv log\left(MRPX_{ijs}\right) - \underbrace{\frac{1}{N}\sum_i log\left(MRPX_{ijs}\right)}_{\equiv log\left(\overline{MRPX}_{ijs}\right)}$$

We can write the covariance of distortions as:

$$Cov\left(\widetilde{\tau}_{is}^R, \widetilde{\tau}_{is}^{mix}\right) = \frac{1}{N}\sum_i \left(\widetilde{\tau}_{is}^{mix} - \widetilde{\tau}_s^{mix}\right)\left(\widetilde{\tau}_{is}^R - \widetilde{\tau}_s^R\right)$$

$$= -\sum_{j\neq L}\alpha_{js}\left(\underbrace{\frac{1}{N}\sum_i \Delta log\left(MRPL_{is}\right)\Delta log\left(MRPX_{ijs}\right)}_{Cov(log(MRPL_s), log(MRPX_{js}))} - \underbrace{\frac{1}{N}\sum_i \left(\Delta log\left(MRPL_{is}\right)\right)^2}_{\sigma^2_{log(MRPL)}}\right) \tag{A47}$$

Which shows that the sign of the covariance term for distortions will depend on the relative size of the covariance between marginal revenue products with the marginal product of labor (average across other inputs) and the variance of log MRPL.

Discussion: The contribution of any distortion type (mix or revenue) to dispersion can only be negative if the covariance above is positive. Otherwise, removal of either distortion type would only lower dispersion. Equation (4) shows that this is likely the case if the covariance between marginal revenue products is relatively low. Alternatively, we could say that a positive covariance between distortions requires a relatively high elasticity between marginal revenue products.

We can more formally evaluate the expression for the contribution of each input type to dispersion (inclusive of covariance). For the input distortion, we have:

$$\sigma^2_{\widetilde{\tau}_s^{mix}} - 2Cov(\widetilde{\tau}_{is}^R, \widetilde{\tau}_{is}^{mix}) = \sum_{j\neq L}\alpha_{js}^2\sigma^2_{log(MRPX_{js})} + \left(\alpha_{Ls}^2 - 1\right)\sigma^2_{log(MRPL_s)} \tag{A48}$$

$$+\sum_{j\neq L}\alpha_{js}\alpha_{Ls}Cov\left(log(MRPX_{ijs}), log(MRPL_{is})\right) + \sum_{j\neq k}\sum_{k\neq j,L}\alpha_{js}\alpha_{ks}Cov\left(log\left(MRPX_{ijs}\right), log\left(MRPX_{iks}\right)\right)$$

The emphasis here now is that $\alpha_{Ls}^2 - 1$ term is negative. Thus, this will tend to create a negative contribution when covariances between marginal revenue products are small or negative, and variance in the normalized variable is small. Likewise, for the revenue distortion, we have:

$$\sigma^2_{\widetilde{\tau}_s^R} - 2Cov(\widetilde{\tau}_{is}^R, \widetilde{\tau}_{is}^{mix}) = (2\alpha_{Ls} - 1)\sigma^2_{log(MRPL_s)} + 2\sum_{j\neq L}\alpha_{js}Cov(log(MRPL_{is}), log(MRPX_{ijs})) \tag{A49}$$

Note here that when the normalized input has a small elasticity, this is more likely to be negative.

A.8.3 Mechanisms under Hsieh and Klenow (2009)

Markdowns Under labor markdowns alone in a two-input production function, covariance between marginal revenue products is zero, and variance in marginal revenue products other than labor is also zero, so:

$$\sigma^2_{\widetilde{\tau}^{mix}} - 2Cov\left(\widetilde{\tau}^R_{is}, \widetilde{\tau}^{mix}_{is}\right) = (\alpha^-_{Ls}1)\sigma^2_{log(MRPL)} < 0. \tag{A50}$$

Likewise, the contribution from revenue distortions is

$$\sigma^2_{\widetilde{\tau}^R_s} - 2Cov(\widetilde{\tau}^R_{is}, \widetilde{\tau}^{mix}_{is}) = (2\alpha_{Ls} - 1)\sigma^2_{log(MRPL_s)}. \tag{A51}$$

This is negative as long as the normalized input has less than 50 percent of the cost share.

Markups Again, normalize the labor margin. Consider that:

$$\frac{MRPX_{ijs}}{MRPL_{is}} = 1$$

$$MRPL_{is} = \mu_{is}$$

. Thus, we have:

$$\sigma^2_{\widetilde{\tau}^{mix}} - 2Cov\left(\widetilde{\tau}^R_{is}, \widetilde{\tau}^{mix}_{is}\right) = 0.$$

$$\sigma^2_{\widetilde{\tau}^R_s} - 2Cov(\widetilde{\tau}^R_{is}, \widetilde{\tau}^{mix}_{is}) = \sigma^2_{log(MRPL_s)}.$$

Thus, we would likely not see the negative contributions we see empirically if markups were the only story.

A.8.4 Bils, Klenow, and Ruane (2021) Normalization

We need to modify our covariance formula, since there is no revenue distortion. Defining $\widetilde{\tau}_{ijs} = log\left(1 + \tau_{ijs}\right)$, we have the following:

$$\sigma^2_{\text{tfpr}} = \sigma^2_{log(\tau)} = \sum_j \alpha^2_{js}\sigma^2_{\widetilde{\tau}_j} + 2\sum_j \sum_{k>j} Cov\left(\widetilde{\tau}_{ijs}, \widetilde{\tau}_{iks}\right).$$

Where $\sigma^2_{\widetilde{\tau}_{js}}$ is the variance of $\widetilde{\tau}_{ijs}$ within sector s.

Under the normalization in Bils, Klenow, and Ruane (2021), we have:

$$1 + \tau_{ijs} = MRPX_{ijs}.$$

Denote the distortion without input k distortion:

$$\tau^{-k}_{is} = \prod_{j\neq k}\left(1 + \tau_{ijs}\right)^{\alpha_{js}} = \prod_{j\neq k} MRPX^{\alpha_{js}}_{ijs}.$$

Taking logs:

$$\widetilde{\tau}^{-k}_{is} = \sum_{j\neq k}\alpha_{js}\widetilde{\tau}_{ijs} = \sum_{j\neq k}\alpha_{js} log\left(MRPX_{ijs}\right).$$

We can then simply plug this into the variance covariance formula above:

$$\sigma^2_{\text{tfpr}} = \sum_j \alpha^2_{js}\sigma^2_{log(MRPX_{ijs})} + 2\sum_j \sum_{k>j} Cov\left(log\left(MRPX_{ijs}\right), log\left(MRPX_{iks}\right)\right).$$

The contribution of input k is now:

$$\alpha^2_{ks}\sigma^2_{log(MRPX_{iks})} + 2\sum_{j\neq k} Cov\left(log\left(MRPX_{ijs}\right), log\left(MRPX_{iks}\right)\right). \tag{A52}$$

Clearly, the contribution is positive if the covariance is positive, and will only be negative when the covariance is substantially negative. This will in turn depend on relative standard deviations of log marginal revenue products, elasticities, and correlations. In the data, we find that the contributions are generally positive.

A.8.5 Mechanisms under BKR Normalization

Markdowns Consider labor markdowns only. In this case, the variance of $logMRPK_{is}$ is zero, and thus there is no variance contribution of the capital distortion or covariance term. Shutting down the labor input distortion yields no dispersion, so its contribution comprises all of the total variance.

Markups As with the Hsieh and Klenow (2009) normalization, we have that $MRPX_{ijs} = MRPK_{iks} = \overline{MRPX}_{is}$ for all plants and all inputs j and k. This means the correlation and variance is the same for all inputs. So the contribution of input k is:

$$\alpha^2_{ks}\sigma^2_{log\left(\overline{MRPX}_{is}\right)} + 2\sigma^2_{log\left(\overline{MRPX}_{is}\right)}\alpha_{ks}\sum_{j\neq k}\alpha_{js} = \alpha_{ks}\sigma^2_{log\left(\overline{MRPX}_{is}\right)}\left(\alpha_{ks} + 2\left(1-\alpha_{ks}\right)\right)$$

$$= \alpha_{ks}\left(2-\alpha_{ks}\right)\sigma^2_{log\left(\overline{MRPX}_{is}\right)}.$$

The first derivative of this expression with respect to α_{ks} is:

$$\frac{d}{d\alpha_{ks}}\alpha_{ks}\sigma^2_{log\left(\overline{MRPX}_{is}\right)} = 2\left(1-\alpha_{ks}\right)\sigma^2_{log\left(\overline{MRPX}_{is}\right)}.$$

This expression is positive for all $\alpha_{ks} < 1$, which is the case by construction. Since these are estimated using cost shares, the larger inputs in terms of costs will account for more of the variance. For example, an input like materials with a cost-share around 0.7, as in Figure (2)(c), will account for 91 percent of the variation.

A.8.6 Mechanisms under our Proposed Normalization

Markdowns Consider labor markdowns only. In this case, contributions will be

$$\sigma^2_{\widetilde{\tau}_{rev}} - 2Cov(\widetilde{\tau}^R_{is}, \widetilde{\tau}^{mix}_{is}) = 2Cov\left(log\left(MRPL^{\alpha_{Ls}}_{is}\right), log\left(\left(1-\alpha_{Ls}+\frac{\alpha_{Ls}}{MRPL_{is}}\right)^{-1}\right)\right) - \sigma^2_{log\left(1-\alpha_{Ls}+\frac{\alpha_{Ls}}{MRPL_{is}}\right)} \tag{A53}$$

and

$$\sigma^2_{\widetilde{\tau}^{mix}_s} - 2Cov(\widetilde{\tau}^R_{is}, \widetilde{\tau}^{mix}_{is}) = Var\left(log\left(MRPL^{\alpha_{Ls}}_{is}\right)\right) - Var\left(log\left(\left(1-\alpha_{Ls}+\frac{\alpha_{Ls}}{MRPL_{is}}\right)^{-1}\right)\right). \tag{A54}$$

Here, a first-order Taylor approximation of the variance of the harmonic mean is used (similar to the delta method in statistics).

$$Var\left(log\left(\left(1-\alpha_{Ls}+\frac{\alpha_{Ls}}{MRPL_{is}}\right)^{-1}\right)\right) = Var\left(log\left(1-\alpha_{Ls}+\alpha_{Ls}exp\left(-log\left(MRPL_{is}\right)\right)\right)\right)$$

$$\approx \left(\frac{\alpha_{Ls}exp\left(-\overline{log(MRPL)}\right)}{1-\alpha_{Ls}+\alpha_{Ls}exp\left(-\overline{log(MRPL)}\right)}\right)^2 Var\left(log\left(MRPL_{is}\right)\right),$$

where $\overline{log(MRPL)} = \frac{1}{N}\sum_i log\left(MRPL_{is}\right)$. This can be simplified to:

$$Var\left(log\left(\left(1-\alpha_{Ls}+\frac{\alpha_{Ls}}{MRPL_{is}}\right)^{-1}\right)\right) \approx \left(\frac{\alpha_{Ls}}{(1-\alpha_{Ls})\,\overline{MRPL}+\alpha_{Ls}}\right)^2 Var\left(log\left(MRPL_{is}\right)\right),$$

where $\overline{MRPL} = exp(\overline{log(MRPL)})$. Likewise, we can approximate the covariance between the log geometric and harmonic means:

$$Cov\left(log\left(MRPL^{\alpha_{Ls}}_{is}\right), log\left(\left(1-\alpha_{Ls}+\frac{\alpha_{Ls}}{MRPL_{is}}\right)^{-1}\right)\right) \approx \frac{\alpha^2_{Ls}}{(1-\alpha_{Ls})\,\overline{MRPL}+\alpha_{Ls}} Var\left(log\left(MRPL_{is}\right)\right)$$

Plugging into our expression for contributions:

$$\sigma^2_{\widetilde{\tau}_{rev}} - 2Cov(\widetilde{\tau}^R_{is}, \widetilde{\tau}^{mix}_{is}) \approx \frac{\alpha^2_{Ls}}{(1-\alpha_{Ls})\,\overline{MRPL}+\alpha_{Ls}}\left(2 - \frac{1}{(1-\alpha_{Ls})\,\overline{MRPL}+\alpha_{Ls}}\right)\sigma^2_{log(MRPL)} \tag{A55}$$

and

$$\sigma^2_{\widetilde{\tau}^{mix}_s} - 2Cov(\widetilde{\tau}^R_{is}, \widetilde{\tau}^{mix}_{is}) \approx \alpha^2_{Ls}\left(1 - \frac{1}{(1-\alpha_{Ls})\,\overline{MRPL}+\alpha_{Ls}}\right)\sigma^2_{log(MRPL)}. \tag{A56}$$

The above implies the following to conditions (which apply for relatively small variance or if other conditions on higher moments are met). First, the contribution of the revenue distortion is (approximately) positive under our normalization so long as $(1-\alpha_{Ls})\,\overline{MRPL}+\alpha_{Ls} > \frac{1}{2}$. To see this, note that since $\overline{MRPL}$ is bounded from below by zero, the largest the second term in parentheses of (A55) can be is 1, and thus less than 2, so the coefficient is always positive. This will hold, for example, when $\overline{MRPL} > 1$. Likewise, the approximate contribution of the mix distortion is positive so long as $\overline{MRPL} > 1$ or, equivalently $\overline{log(MRPL)} > 0$. One can see immediately that when $\overline{MRPL} = 1$, the contribution is zero. Under what conditions will $\overline{MRPX} > 1$? Recall the definition:

$$MRPX_{is} = \rho_s \alpha_{js} \frac{R_{is}}{w_{js}X_{ijs}}.$$

From our identification strategy, we have:

$$\rho_s = \frac{\sum_i \sum_j w_{js} X_{ijs}}{\sum_i R_{is}}$$

and

$$\alpha_{js} = \frac{\sum_i w_{js} X_{ijs}}{\sum_i \sum_k w_{ks} X_{iks}}.$$

So, we have:

$$\rho_s \alpha_{js} = \frac{\sum_i w_{js} X_{ijs}}{\sum_i R_{is}} = \left(\sum_i \frac{w_{js} X_{ijs}}{\sum_{i_1} w_{js} X_{i_1 js}} \frac{R_{is}}{w_{js} X_{ijs}} \right)^{-1},$$

and therefore (defining $X_{js} \equiv \sum_i X_{ijs}$):

$$MRPX_{is} = \frac{\frac{R_{is}}{w_{js} X_{ijs}}}{\sum_{i_1} \frac{w_{js} X_{i_1 js}}{w_{js} X_{js}} \frac{R_{i_1 s}}{w_{js} X_{i_1 js}}}.$$

In words, this value is greater than one when the average revenue product $\frac{R_{is}}{w_{js} X_{ijs}}$ is greater than the weighted mean of the average marginal revenue product. Taking logs, we have:

$$log\left(MRPX_{is}\right) = log\left(\frac{R_{is}}{w_{js} X_{ijs}} \right) - log\left(\sum_{i_1} \frac{w_{js} X_{i_1 js}}{w_{js} X_{js}} \frac{R_{i_1 s}}{w_{js} X_{i_1 js}} \right).$$

Again, this is greater than zero so long as the average revenue product is greater than the weighted mean across plants. Taking an average to obtain $\overline{log(MRPX)}$ we have:

$$\overline{log(MRPX)} = \frac{1}{N} \sum_i log\left(\frac{R_{is}}{w_{js} X_{ijs}} \right) - log\left(\frac{R_s}{w_{js} X_{js}} \right).$$

If the unweighted mean of log average revenue products is greater than the log of the weighted mean, then this will hold. This would be the case if relatively large plants had relatively low average revenue products compared to the typical plant.

Markups As with the Hsieh and Klenow (2009) normalization, we have that $MRPX_{ijs} = MRPK_{iks}$ for all plants and all inputs j and k. So the following relationship holds:

$$\prod_j MRPX_{ijs}^{\alpha_{js}} = \left(\sum_j \frac{\alpha_{js}}{MRPX_{ijs}} \right)^{-1}.$$

Thus our mix distortion has no variation:

$$\prod_j \left(1 + \tau_{ijs}\right)^{\alpha_{js}} = \frac{\prod_j MRPX_{ijs}^{\alpha_{js}}}{\left(\sum_j \frac{\alpha_{js}}{MRPX_{ijs}} \right)^{-1}} = 1.$$

So the contribution of the mix distortion is zero, and the log TFPR dispersion is entirely described by the margin distortion.

A.9 Responsiveness

A different way to consider distinct margin and mix distortions is to consider responsiveness of inputs to changes in TFPQ. The input demand equation (see Appendix A.6) implies that in the absence of distortions that are correlated with TFPQ, all inputs will have the same elasticity of response to TFPQ. However, in the context of correlated distortions with TFPQ, differential responsiveness can be informative. If, for example, the only distortion is markups, and markups increase with size (which is increasing in TFPQ), then responsiveness of all inputs will remain the same with respect to TFPQ. Moreover, if markups are rising on average over time, responsiveness for all inputs will decline. Alternatively, if factor specific distortions are correlated with TFPQ then there can be differential responsiveness in any given year and changes in responsiveness over time. Since for example, the factor specific distortions may be place holders for adjustment costs (see, discussion in Decker et al., 2016), such distortions are inherently correlated with TFPQ. The reason is that factors subject to greater adjustment costs will be less responsive to TFPQ. Moreover, if adjustment costs are rising for some factors relative to others over time, such factors will become increasingly less responsive over time to TFPQ compared to other factors.

With the above considerations in mind, we focus on differential responsiveness patterns across inputs. We investigate responsiveness of the two most important production inputs: employment and intermediate inputs. Specifically, we look for evidence of differential dynamics between these inputs.

A.9.1 Empirical Estimates of Responsiveness

We build on the approach in Decker et al. (2020). For labor and materials inputs, we regress the the DHS growth rate for continuing establishments, defined as $\frac{X_{t+1}-X_t}{0.5(X_{t+1}+X_t)}$, on a measure of TFPQ and control variables. The coefficient of productivity from this regression can then be interpreted as measure of responsiveness.

Figure A3 shows the responsiveness of continuing establishments' employment (from the LBD) and intermediate inputs (from the ASM) to $\ln \widehat{A}_{it}$.[26] The estimated coefficients in the 1980s are similar and have overlapping confidence intervals. However, from that point, responsiveness patterns diverge. Importantly, intermediate input responsiveness declines only slowly while employment responsiveness declines significantly. The implication is that changes in $\widehat{A}_{it}$ are associated with changing input mix, especially beginning in the 2000's.

We interpret the changing responsiveness patterns as a sign that it is not only changes in margin distortions that are at work. Candidates for such mechanisms include adjustment costs rising specifically for labor.

[26] We focus on continuers because measuring entry and exit in the ASM is problematic. We prefer using LBD employment for dynamic analysis, since it is better suited for longitudinal links, and the March 12 timing is more consistent with the initial impact of a productivity shock on subsequent employment relative to the full-year measure from the ASM.



Figure A3: Input responsiveness.

Source: Authors' calculations using CMP and LBD data from continuing establishments, see Section 2. The bars denote the coefficient of productivity from regressing $\frac{X_{it+1}-X_{it}}{0.5(X_{it+1}+X_{it})}$ on $\ln \widehat{A}_{it}$, $\ln X_{it}$ and all interactions of NAICS6 and year dummies. Regressions are sample-weighted, red lines show 95% confidence bands using robust standard errors. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7526913: CBDRB-FY24-CED006-0001, CBDRB-FY24-CED006-0004, CBDRB-FY25-CED006-005, CBDRB-FY25-CED006-015)

A.9.2 Alternative Responsiveness Measures

The series in Figure A3 show alternative measures of responsiveness, and shows the differences between our preferred employment measure for dynamic exercises (LBD employment) and the ASM measure of employment (TE - Total Employment). Figure A4 includes the previous decade. Estimates from the 1970s are less precise due a truncated sample: only 4 years of data is available because the LBD begins in 1976.

A.9.3 Illustrative example: Responsiveness and adjustment costs

The results in this Appendix are intended to illustrate the behavior of a responsiveness measure that is based on a calibrated model of costly input adjustment. The model, described in Bloom (2009), is useful because measured responsiveness follows directly from adjustment costs parameters. The baseline parameter vector, denoted by θ_B, is taken from Bloom (2009). The graphs depict the inaction region under different θs. For each simulation, we calculated the inverse of the approximate area of the inaction region, denoted by $\mathcal{R}=1/\widehat{IA}$, where IA is short for "inaction area" and the approximation is based on the trapezoid rule. The scenarios shown in the panels of Figure A5 illustrate the extent to which an increase in adjustment cost parameters yield a larger inaction region and therefore lower responsiveness. This is highlighted by the substantial decrease in $\mathcal{R}$ as cost parameters increase. The black dashed line in Figure A5a shows the inaction region when uncertainty increases, i.e., at the beginning of a downturn. The region expands and $\mathcal{R}$ drops from 0.22 to 0.13. This is a useful benchmark for evaluating the effect of cost parameters because increasing uncertainty can mark the beginning of a downturn, which is not an unusual event. The orange line shows results after increasing adjustment cost parameters by one standard

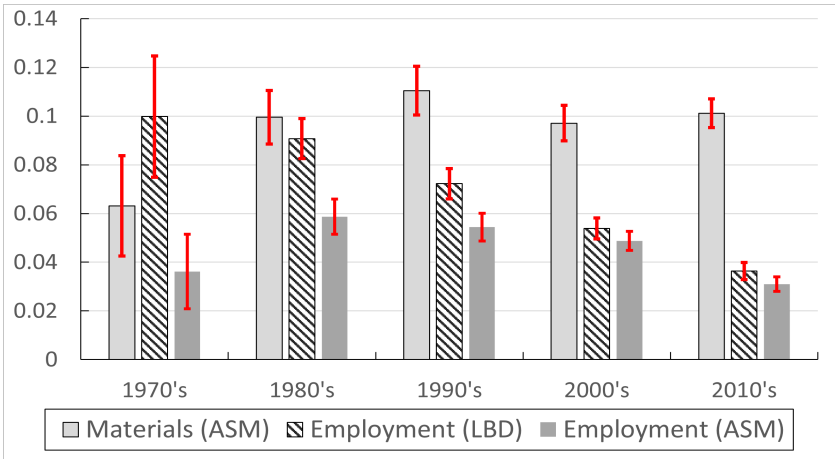
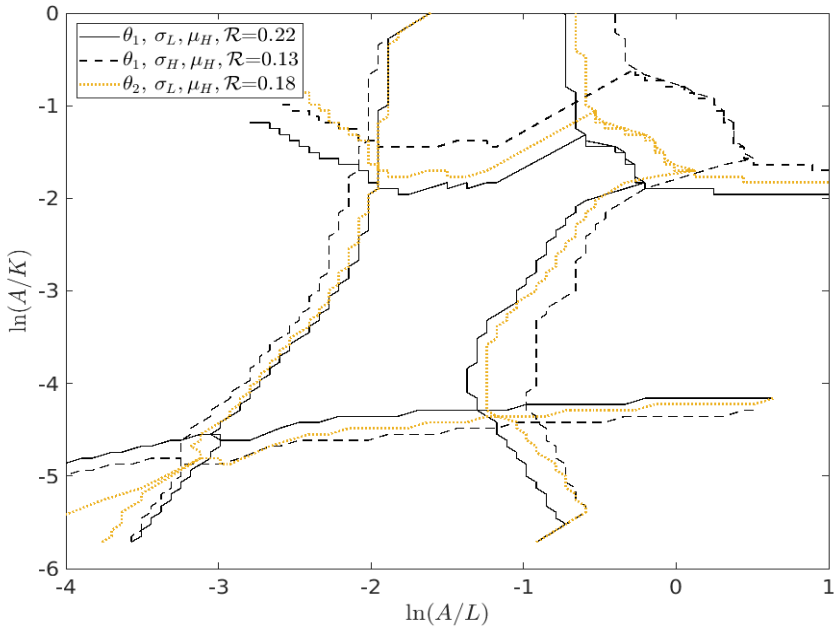


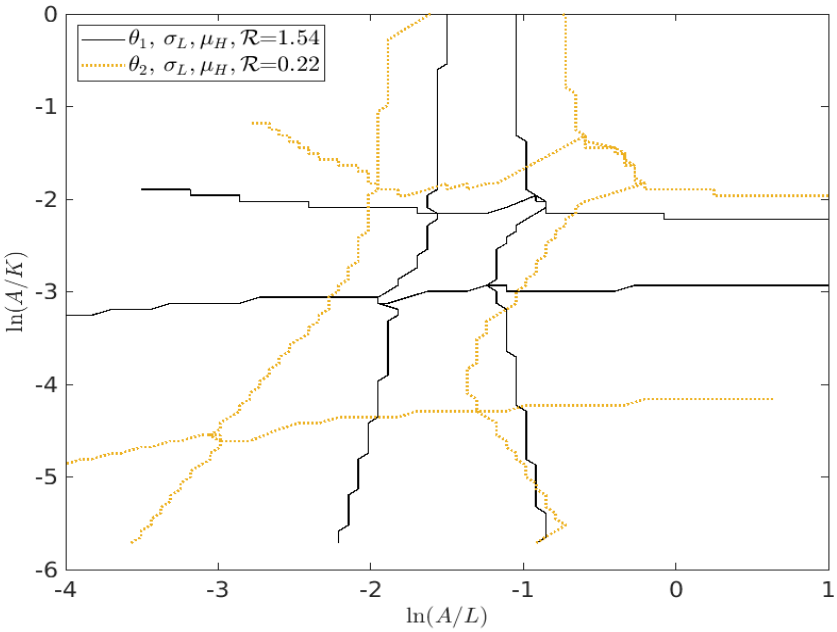
Figure A4: Alternative Responsiveness.

Source: Authors' calculations using CMP data, see Section A.9. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7526913: CBDRB-FY24-CED006-0001, CBDRB-FY24-CED006-0004, CBDRB-FY25-CED006-005, CBDRB-FY25-CED006-015).

error holding everything else constant. $\mathcal{R}$ declines, which illustrates that a moderate adjustment cost increase leads to a decline in $\mathcal{R}$. On the other hand, the effect is smaller than the effect of an increase in uncertainty, which can be interpreted as an indication that typical adjustment cost changes may not be sufficient to entail more significant declines in $\mathcal{R}$. Figure A5b shows a scenario where adjustment costs are increased from a near-zero range (1/10 of the baseline) to the baseline, which yields a significant increase in the inaction area and a drop in $\mathcal{R}$.



(a) Impact of uncertainty and a one-standard-deviation increase in θ.



(b) Impact of increasing θ from near zero.

Figure A5: The effect of adjustment cost parameters on $\mathcal{R}$.

Source: Authors' simulations using the model described in Bloom (2009), based on a 100-point grid and zero quadratic costs. Responsiveness is denoted by $\mathcal{R}$ and is defined as $\mathcal{R}=1/\widehat{IA}$ where $\widehat{IA}$ denotes the area of the inaction region approximated by the trapezoid rule. Panel A5a illustrates the effect of an increase in uncertainty (dashed black line) and a one-standard-error increase in the estimated adjustment cost parameter vector (dotted orange line): $\theta_2=\theta_1(1+\text{se}(\theta_1))$ where $\theta_1=\theta_B=(C_K^F, C_L^F, C_K^P, C_L^P)$ =(0.0146, 0.0208, 0.3392, 0.0181). Panel A5b shows the effect of changing $\theta_1=0.1\times\theta_B$ to $\theta_2=\theta_B$.

A.10 Quantity based measurement of inputs

We measure input variables using expenditures in data divided by industry deflators. This is appropriate if the establishment specific input price variation reflects quality differences. However, it may be that such variation reflects other factors. In this section, we explore the sensitivity of our results to this issue focusing on the labor input. We exploit the availability of total hours to shed some light on the quantitative importance of these issues. Reported hours data yields an alternative $\widehat{A}_{is}$.

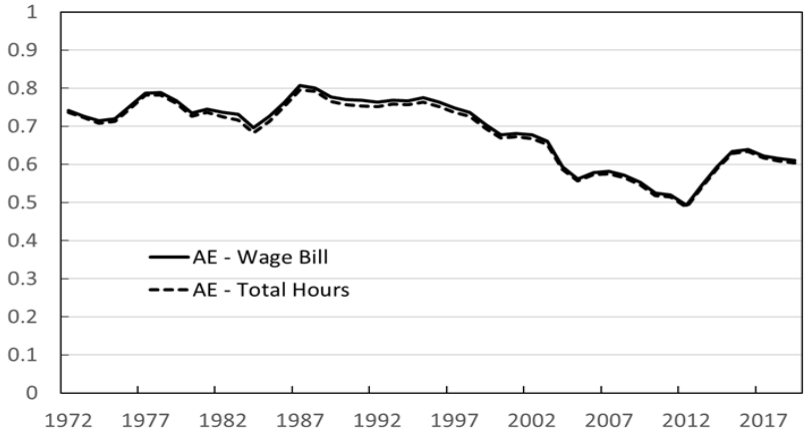


Figure A6: Allocative efficiency under alternative labor input measures

Source: Authors' calculations using CMP data with a three-year moving average, see Section 2. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7526913: CBDRB-FY24-CED006-0001, CBDRB-FY24-CED006-0004, CBDRB-FY25-CED006-005, CBDRB-FY25-CED006-015).

Figure A6 displays allocative efficiency under the two labor input measures. The solid line shows the headline series where distortions and $\widehat{A}_{is}$ are calculated using expenditure data, and hours are derived from deflating plant-level total salaries with the aggregate wage. The dashed line shows the case for reported hours data. The relatively small difference between the two series indicates that such a distinction has a small effect on measured allocative efficiency. Although the difference could be larger if we considered the entire input index.[27]

A.11 Normality of the within-industry productivity distribution

Existing evidence suggests that the within-industry productivity distribution has long right tails, see Cunningham et al. (2023). The exercise below is based on the idea that for a standard normal distribution the range r_1=6σ covers approximately 99.7% of observations. Furthermore, the difference between the 1st and 99th percentiles, denoted by r_2, corresponds to 4.65σ under normality. Therefore, under the null hypothesis of a log-normal distribution the expected difference $r_2 - r_1$ is $4.65\sigma - 6\sigma = -1.35\sigma$. Since $0 < \sigma$, the difference under the null hypothesis should be

[27]Labor is the second most important factor, see Figure 2a.

negative. Using dispersion statistics from DiSP[28], we find that the mean of $r_2 - r_1$ is approximately +0.5, which implies the empirical distribution exhibits fat tails compared to a normal distribution. As an illustration Figure A7 plots the cross-industry variation of r_2-r_1 in DiSP, along with a normal density with the same μ, σ and N and a normal density with N=10000.

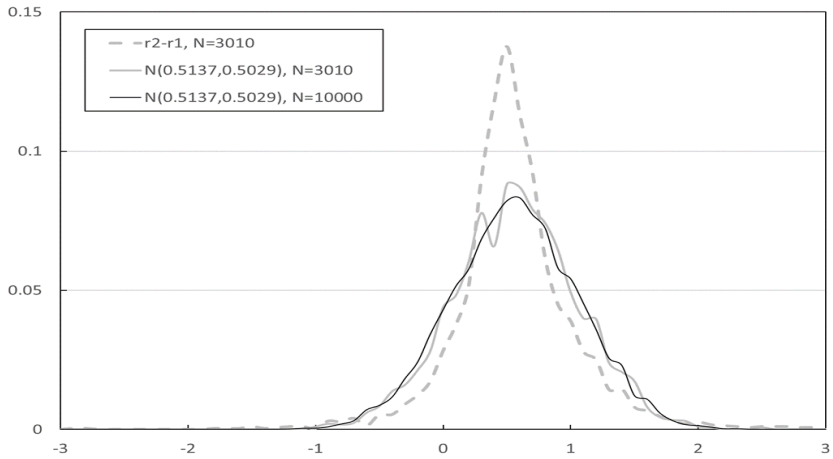


Figure A7: Estimated probability densities: $r_2 - r_1$ and two normal random variables

Source: Authors' calculations using DiSP data. The gray dashed line shows the estimated density of r_2-r_1, where r_2=$d101$+$d9010$+$d9990$, r_1=6σ are taken from DiSP. The solid gray line shows the density of a normal random variable with the same μ, σ and N as the empirical distribution. The solid black line shows shows the density function of a normal random variable with N=10000.

[28]Dispersion Statistics on Productivity is an experimental data product of the US Census Bureau.